

# Variational methods for partial differential equations

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January 2024

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# 1 Elements of functional analysis

## 1.1 Banach spaces

**Definition 1.1.** (Norms on a vector space). Let  $V$  be a vector space over a field  $\mathbb{F}$ . A norm  $\|\cdot\|$  on  $V$  is a function from  $V$  to  $\mathbb{R}_{\geq 0} = [0, +\infty)$  that satisfies the following three properties:

1.  $\|u + v\| \leq \|u\| + \|v\|$  for all  $u, v \in V$ .
2.  $\|\lambda u\| \leq |\lambda| \|u\|$  for all  $u \in V$  and  $\lambda \in \mathbb{F}$ .
3.  $\|u\| = 0$  if and only if  $u = 0$ .

**Definition 1.2.** (Semi-norms) If a function  $\|\cdot\|$  from  $V$  to  $\mathbb{R}_{\geq 0}$  satisfies properties 1,2 but not 3, then we say that  $\|\cdot\|$  is a semi-norm.

**Example.** (Norms on  $\mathbb{R}^n$ ). The Euclidean norm is  $\|x\| = (\sum_{j=1}^n |x_j|^2)^{1/2}$  where  $x \in (x_1, \dots, x_n) \in \mathbb{R}^n$ .

We can check if it satisfies three axioms:

1.

$$\begin{aligned}\|x + y\| &= \left( \sum_{j=1}^n |x_j + y_j|^2 \right)^{1/2} \\ \|x + y\|^2 &= \left( \sum_{j=1}^n |x_j|^2 \right)^{1/2} + \left( \sum_{j=1}^n |y_j|^2 \right)^{1/2} + \left( \sum_{j=1}^n 2|x_j y_j| \right)^{1/2} \\ \|x + y\|^2 &\leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \\ \|x + y\| &\leq \|x\| + \|y\|\end{aligned}$$

where Cauchy-Schwarz inequality is used.

2.

$$\begin{aligned}\|\lambda x\| &= \left( \sum_{j=1}^n |\lambda x_j|^2 \right)^{1/2} \\ &= \left( \sum_{j=1}^n |x_j|^2 |\lambda|^2 \right)^{1/2} = |\lambda| \|x\|\end{aligned}$$

3. Trivial since all  $|x_j|$  are non-negative.

**Example.** More generally, for any real number  $1 \leq p < \infty$ , the  $p$ -norm on  $\mathbb{R}^n$  is defined by  $\|x\|_p = (\sum_{j=1}^n |x_j|^p)^{1/p}$ . The  $\infty$ -norm  $\|\cdot\|$  is defined by  $\|x\|_\infty = \max_{1 \leq j \leq n} |x_j|$ .

**Definition 1.3.** (Equivalence of norms) we say that two norms  $\|\cdot\|_1$  and  $\|\cdot\|_2$  are equivalent if there exists positive constants  $c_1 > 0$  and  $c_2 > 0$  such that

$$c_1 \|v\|_1 \leq \|v\|_2 \leq c_2 \|v\|_1 \quad \forall v \in V$$

**Example.** Show that  $\|\cdot\|_p$  and  $\|\cdot\|_1$  are equivalent on  $\mathbb{R}^n$ .

**Proof.** If the norm is not finite, there are nothing to show. Now assuming it's finite then it suffices to show that there exists some constants  $C_1, C_2$  such that

$$C_1 \|x\|_1 \leq \|x\|_p \leq C_2 \|x\|_1$$

for all  $x \in \mathbb{R}^n$ . This is trivial if  $x = 0$  so we assume  $x \neq 0$ , in which case we can divide by  $\|x\|_1$  to obtain

$$C_1 \leq \|u\|_p \leq C_2$$

where  $u = x/\|x\|_1$  has norm  $\|u\|_1 = 1$ . □

**Definition 1.4.** A sequence  $\{v_n\}$  is called a Cauchy sequence if, for all  $\varepsilon > 0$ , there exists a number  $N \geq 1$  such that  $\|v_m - v_n\| < \varepsilon$  for all  $n, m \geq N$ .

**Definition 1.5.** A normed space  $(V, \|\cdot\|)$  is called a Banach space if  $(V, \|\cdot\|)$  is complete: every Cauchy sequence has a limit in  $V$ .

**Examples.**

1.  $\mathbb{R}^n$  with any norm  $(\|\cdot\|, \|\cdot\|_p, \dots)$  is a Banach space.

2.  $\ell_p$  ( $p \in [1, \infty]$ ) are Banach spaces
3. The space  $C[a, b]$  of real valued functions on a non-empty interval  $[a, b]$  that are uniformly continuous on  $(a, b)$  (equivalent to continuous on  $[a, b]$ ), with the norm  $\|f\| = \max_{x \in [a, b]} |f(x)|$ . Then  $C[a, b]$  is a Banach space.

**Example.**  $C[a, b]$  with the  $L^2[a, b]$  norm

$$\|f\|_{L^2(a, b)} = \left( \int_a^b |f(x)|^2 dx \right)^{1/2}$$

is not a Banach space.

**Theorem 1.6.** (Banach Fixed Point Theorem) Let  $(V, \|\cdot\|)$  be a Banach space, and let  $T : V \rightarrow V$  be a contraction mapping i.e.  $\exists \rho \in [0, 1)$  such that

$$\|T(u) - T(v)\| \leq \rho \|u - v\| \quad \forall u, v \in V$$

then  $T$  has a unique fixed point in  $V$  : there exists a unique  $u^* \in V$  such that  $T(u^*) = u^*$ .

**Proof.** Choose  $v_0 \in V$  define  $(v_n)$

$$v_{n+1} = T(v_n)$$

then

$$\begin{aligned} d(v_{m+1}, v_m) &= d(T(v_m), T(v_{m-1})) \\ &\leq \rho d(v_m, v_{m-1}) \leq \rho^m d(v_1, v_0) \end{aligned}$$

now, say  $n > m$

$$\begin{aligned} d(v_m, v_n) &\leq d(v_m, v_{m+1}) + \dots + d(v_{n-1}, v_n) \\ &= (\rho^m + \dots + \rho^{n-1}) d(v_1, v_0) \\ &= \rho^m \left( \frac{1 - \rho^{n-m}}{1 - \rho} \right) d(v_1, v_0) \end{aligned}$$

since  $0 \leq \rho < 1$  we have

$$d(v_m, v_n) \leq \frac{\rho^m}{1 - \rho} d(v_1, v_0)$$

since  $V$  is a Banach space so it is complete as well in this case  $\exists v \in V$  such that  $v_n \rightarrow v$ . Thus,

$$\begin{aligned} d(v, T(v)) &\leq d(v, v_n) + d(v_n, T(v)) \\ &= d(v, v_n) + d(T(v_{n-1}), T(v)) \\ &\leq d(v, v_n) + \rho d(v_{n-1}, v) \end{aligned}$$

we can pick  $n$  to be large in that case  $d(v, v_n), d(v_{n-1}, v) \rightarrow 0$ . Thus,  $d(v, T(v)) = 0 \implies T(v) = v$ . Now we only need to show the uniqueness, given  $v = T(v)$  and  $\tilde{v} = T(\tilde{v})$  then

$$d(v, \tilde{v}) = d(T(v), T(\tilde{v})) \leq \rho d(v, \tilde{v})$$

However, since  $\rho < 1$  then  $d(v, \tilde{v}) = 0 \implies v = \tilde{v}$ . □

**Definition 1.7.** (Bounded Linear Operators) For normed spaces  $V, W$ , a linear operator  $T : V \rightarrow W$  is called bounded if and only if

$$\|T\|_{\mathcal{L}[V, W]} = \sup_{v \in V, \|v\|_V \leq 1} \|Tv\|_W < \infty$$

$\|T\|_{\mathcal{L}[V, W]}$  is the space of all Bounded Linear Operators from  $V \rightarrow W$ . It is a normed space with norm  $\|\cdot\|_{\mathcal{L}[V, W]}$ .

**Example.** Show that the following are equivalent ( $T : V \rightarrow W$  is linear):

1.  $T : V \rightarrow W$  is a Bounded Linear Operators.
2.  $\exists M < \infty$  such that  $\|Tv\|_W \leq M\|v\|_V \quad \forall v \in V$ .
3.  $T$  is a continuous everywhere on  $V$ .

**Proof.** We first show that  $1 \implies 2$ . If  $T$  is bounded, let  $v \in V$ . If  $v = 0$  then  $Tv = 0$  then there is nothing to show. If  $v \neq 0$ , now define  $\hat{v} = \frac{v}{\|v\|_V}$  then  $\|\hat{v}\|_V = 1$ . So by definition

$$\|T\hat{v}\|_W \leq \sup_{\hat{v} \in V, \|\hat{v}\|_V \leq 1} \|T\hat{v}\|_W = \|T\|_{\mathcal{L}[V, W]}$$

$$\implies \frac{1}{\|v\|_V} \|Tv\|_W \leq \|T\|_{\mathcal{L}[V, W]} \implies \|Tv\|_W \leq \|T\|_{\mathcal{L}[V, W]} \|v\|_V$$

then  $M = \|T\|_{\mathcal{L}[V, W]}$ . Now we show that  $2 \implies 1$ : If  $\exists n < \infty$  such that  $\|Tv\|_W \leq n\|v\|_V$

$$\implies \sup_{v \in V, \|v\|_V \leq 1} \|Tv\|_W \leq n \iff \|T\|_{\mathcal{L}[V, W]} \leq n$$

and  $T$  is bounded. □

**Proposition 1.8.** If  $W$  is a Banach space, then  $\mathcal{L}(V, W)$  is also a Banach space.

**Proof.** We first show that  $\mathcal{L}[V, W]$  is complete. By definition,

$$\|T\|_{\mathcal{L}[V, W]} = \sup_{v \in V, \|v\|_V = 1} \|Tv\|_W$$

for each  $x \in V$ ,  $T_n x$  is Cauchy in  $W$  then  $T_n x$  converges to some  $Tx$  which implies  $\|T\|_{\mathcal{L}[V, W]} \leq M$  for some constant  $M$ . For all  $n$  and  $x \in V$ ,

$$\begin{aligned} \|Tx\|_W &\leq \|Tx - T_n x\|_W + \|T_n x\|_W \\ &\leq \|Tx - T_n x\|_W + \|T_n\|_{\mathcal{L}[V, W]} \|x\|_V \end{aligned}$$

take  $n \rightarrow \infty$  we have  $\|Tx\|_W \leq M\|x\|_V$  so  $T \in \mathcal{L}[V, W]$  with  $\|T\|_{\mathcal{L}[V, W]} \leq M$  so it is complete. Now we need to show that  $T_n \rightarrow T$  in  $\mathcal{L}[V, W]$ . Given  $\varepsilon > 0$ , we choose  $N$  such that  $\|T_n - T_m\|_{\mathcal{L}[V, W]} < \varepsilon/2$ . We prove the following claim: For each  $x \in V$  with  $x \neq 0$  and  $n \geq N$  we have  $\|T_n x - Tx\|_W \leq \varepsilon\|x\|_V$ . Let  $n \geq N$  since  $T_m x \in Tx$  in  $W$  so we are able to choose  $m(x) \geq N$  such that

$$\|T_{m(x)} x - Tx\|_W \leq \frac{\varepsilon}{4} \|x\|_V$$

then

$$\begin{aligned} \|Tx - T_n x\|_W &= \|Tx - T_{m(x)} x + T_{m(x)} x - T_n x\|_W \\ &\leq \|Tx - T_{m(x)} x\|_W + \|T_{m(x)} x - T_n x\|_W \\ &\leq \frac{\varepsilon}{4} \|x\|_V + \|T_{m(x)} - T_n\|_{\mathcal{L}[V, W]} \|x\|_V \\ &\leq \frac{\varepsilon}{4} \|x\|_V + \frac{\varepsilon}{2} \|x\|_V < \varepsilon \|x\|_V \end{aligned}$$

Hence  $\|T - T_n\|_{\mathcal{L}[V, W]} < \varepsilon$  for  $n \geq N$ . This shows that  $T_n \rightarrow T$  in  $\mathcal{L}[V, W]$ .  $\square$

**Definition 1.9.** A Bounded Linear Operators from  $V \rightarrow \mathbb{R}$  is called a bounded linear functional on  $V$  (BLF). The dual space  $V^*$  is the space of all BLF on  $V$  ( $V^* = \mathcal{L}(V, \mathbb{R})$ ).  $V^*$  is a Banach space with norm

$$\|\psi\|_{V^*} = \sup_{v \in V, \|v\|_V \leq 1} |\psi(v)|$$

**Example.**  $C[a, b] : \text{consider } \psi : C[a, b] \rightarrow \mathbb{R}$

$$\psi(f) = \int_a^b f(x) dx$$

show that it's a bounded linear functional.

**Proof.**

1. Easy to show it's linear.
2. To show that  $\phi$  is bounded, need to show  $\exists M < \infty$  such that  $|\psi(f)| \leq M\|f\|_{C[a, b]}$  note that

$$|\psi(f)| = \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \leq \max_{x \in [a, b]} |f(x)| (b - a) = (b - a) \|f\|_{C[a, b]}$$

$$\begin{aligned} \implies M &= b - a \text{ so } \psi \text{ is bounded. Let } f(x) = 1, \forall x \in [a, b] \implies \psi(f) = \int_a^b 1 dx = b - a \text{ and } \|f\|_{C[a, b]} = 1 \text{ thus} \\ \implies \|\psi\|_* &= \sup_{\|f\|_{C[a, b]} \leq 1} |\psi(f)| \geq b - a \implies \|\psi\|_* = b - a \end{aligned}$$

$\square$

**Definition 1.10.** (Kernel of a BLF) Let  $\psi \in V^*$ , the kernel of  $\psi$  is defined as

$$\ker \psi = \{v \in V \text{ such that } \psi(v) = 0\}$$

$\ker \psi$  is a linear subspace of  $V$ .

**Exercise.** Show that  $\ker \psi$  is closed for all  $\psi \in V^*$ .

**Proof.** By previous example, we know that for a bounded linear operators it's also continuous everywhere as well. Thus, note that  $\ker \psi = \psi^{-1}(\{0\})$  where  $\psi^{-1}$  is continuous and  $\{0\}$  is closed in terminology. Thus,  $\ker \psi$  is closed.  $\square$

## 1.2 Hilbert spaces

**Definition 1.11.** Let  $V$  be a real vector space. An inner product  $(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$  satisfies

- (a) (Linearity)  $(\lambda u + w, v) = \lambda(u, v) + (w, v) \forall \lambda \in \mathbb{R}, u, v, w \in V$ .
- (b) (Symmetry)  $(u, v) = (v, u) \forall u, v \in V$ .
- (c) (Positive definiteness)  $(v, v) > 0$  if  $v \neq 0$ .

Note: (a) + (b) implies  $(\cdot, \cdot)$  bilinear: linear in both arguments, (a) implies  $(v, v) = 0$  if  $v = 0$ . Norm induced by an inner product: given an inner product  $(\cdot, \cdot)$ , the induced norm  $\|\cdot\|$  is defined by  $\|v\| = \sqrt{(v, v)}$  for all  $v \in V$ .

**Proposition 1.12.** (Cauchy-Schwarz inequality)  $|(v, w)| \leq \|v\| \|w\| \forall v, w \in V$ .

**Proof.** In the problem sheet 1. □

**Definition 1.13.** An inner product space  $V$  with inner product  $(\cdot, \cdot)$  is called a Hilbert space if  $V$  is complete with respect to the induced norm  $\|\cdot\|$ .

**Example.** The space  $\ell_2$  has norm  $\|v\|_{\ell_2} = (\sum_{j=1}^{\infty} v_j^2)^{1/2}$  for  $v = \{v_j\}_{j=1}^{\infty}$ . This norm is induced by the inner product

$$(v, w)_{\ell_2} = \sum_{j=1}^{\infty} v_j w_j$$

so  $\ell_2$  is a Hilbert space.

**Definition 1.14.** (Extended real valued)  $f : (a, b) \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$

**Example.** The space  $L^2(a, b)$  is the space of extended real valued functions on  $(a, b)$  that are Lebesgue measurable and square integrable on  $(a, b)$

$$\int_a^b |f(x)|^2 dx < \infty$$

it is a Hilbert space when equipped with the inner product

$$(f, g)_{L^2(a, b)} = \int_a^b f(x)g(x) dx$$

$\forall f, g \in L^2(a, b)$ .

(Closed subspace and orthogonal complements) In finite dimensional spaces, we are used to several properties:

1. Every subspace is closed
2. Every subspace of an inner product space has an orthogonal complement.

In infinite dimensional space, these are no longer true in general. For instance, if  $V$  is a Hilbert space, and  $W$  is a subspace of  $V$ , we can always define the orthogonal set  $W^\perp = \{v \in V : (v, w) = 0 \forall w \in W\}$ . It is possible that  $V \neq W \oplus W^\perp$ .

**Theorem 1.15.** Let  $V$  be a Hilbert space, and let  $W$  be a closed subspace of  $V$ . Then we have

$$V = W \oplus W^\perp$$

then we call  $W^\perp$  the orthogonal complement of  $W$ .

**Proof.** Problem sheet 1. □

**Lemma 1.16.** Let  $V$  a Hilbert space and a  $W$  be a subspace of  $V$ . Then  $V = W$  if and only if  $W$  is closed in  $V$  and  $W^\perp = \{0\}$ .

**Proof.**

( $\implies$ ) then  $W$  is closed by definition, if  $v \in W^\perp$  then  $v \in W \implies v = 0$  so  $W^\perp = \{0\}$ .

( $\impliedby$ ) by Theorem 1.15,  $V = W \oplus W^\perp = W \oplus \{0\} = W$ . □

## 1.3 Riesz Representation Theorem

In a Hilbert space  $V$ , every element of  $V$  can be associated to an element of  $V^*$  by defining  $u \mapsto \psi(v) = (u, v), \forall v \in V$

**Exercise.** Show that  $\|\psi\|_* = \|u\|$ .

**Proof.** It suffices to show that

$$\|u\| = \sup_{v \in V} \frac{|(u, v)|}{\|v\|}$$

first observe that

$$\|u\| = \frac{(u, u)}{\|u\|} \leq \sup_{v \in V} \frac{|(u, v)|}{\|v\|}$$

so we have shown that

$$\|u\| \leq \sup_{v \in V} \frac{|(u, v)|}{\|v\|}$$

Now  $\forall v \in V$ , by Cauchy-Schwarz inequality,  $\|u\| \|v\| \geq |(u, v)|$  which implies

$$\|u\| \geq \sup_{v \in V} \frac{|(u, v)|}{\|v\|}$$

in this case we have

$$\|u\| = \sup_{v \in V} \frac{|(u, v)|}{\|v\|}$$

□

**Theorem 1.17.** (Riesz Representation Theorem) For every  $\psi \in V^*$ ,  $\exists! u \in V$  such that  $(u, v) = \psi(v), \forall v \in V$ . Furthermore  $\|u\| = \|\psi\|_*$ .

**Proof.** Let  $\psi \in V^*$ , define  $W = \ker \psi$ . Then  $W$  is a closed subspace of  $V$ . If  $\psi = 0$  then  $u = 0 \implies u$  solves  $(u, v) = \psi(v), \forall v \in V$ . If  $\psi \neq 0 \implies W \neq V$  ( $\psi(v) \neq 0$  for some  $v \in V$ ). By Lemma 1.16,  $W^\perp \neq \{0\}$ . Therefore, there exists  $u_0 \neq 0$  and  $u_0 \in W^\perp$ . Define

$$u = \frac{\psi(u_0)}{\|u_0\|^2} u_0$$

we claim that this is the solution. To do that we first that  $\psi(u) \neq 0$ . Since  $u \neq 0, u \in W^\perp$ . In addition,  $\|u\| = \frac{|\psi(u_0)|}{\|u_0\|}$ ,  $\psi(u) = \frac{|\psi(u_0)|^2}{\|u_0\|^2}$  so  $\psi(u) = \|u\|^2$ . Let  $v \in V$ , then

$$\psi(v - \frac{\psi(v)}{\psi(u)}u) = \psi(v) - \frac{\psi(v)}{\psi(u)}\psi(u) = 0$$

so  $v - \frac{\psi(v)}{\psi(u)}u \in W = \ker \psi$ . Thus,

$$0 = (u, v - \frac{\psi(v)}{\psi(u)}u) = (u, v) - \frac{\psi(v)}{\psi(u)}(u, u) = (u, v) - \psi(v) \frac{\|u\|}{\psi(u)} = (u, v) - \psi(v)$$

then  $(u, v) = \psi(v)$  which means  $u$  solves  $(u, v) = \psi(v), \forall v \in V$ . Now we only need to show the uniqueness. If  $\hat{u} \in V$  that is also a solution and note that

$$(\hat{u} - u, v) = \psi(\hat{u}) - \psi(u) = 0$$

put  $v = u - \hat{u}$  then

$$\|u - \hat{u}\|^2 = (u - \hat{u}, u - \hat{u}) = 0$$

so  $u = \hat{u}$  so  $u$  is unique. □

**Definition 1.18.** (The Riesz map) The Riesz map is the bounded linear operator  $R$  from  $V^*$  to  $V$  that maps  $\psi \in V^*$  to  $R\psi = u$  where  $u \in V$  is the unique element of  $V$  solving  $(u, v) = \psi(v)$  for all  $v \in V$ .

**Exercise.** Show that  $R$  is linear. Show that  $R$  is an isometry  $\|R\psi\| = \|\psi\|_*$ .

**Proof.** To show it's linear say  $R\psi_1 = u_1$  and  $R\psi_2 = u_2$  the existence of  $u_1$  and  $u_2$  follows by Riesz Representation theorem. Now, for  $a, b \in \mathbb{F}$  we have

$$(au_1 + bu_2, v) = a(u_1, v) + b(u_2, v) = a\psi_1(v) + b\psi_2(v) = (a\psi_1 + b\psi_2)(v)$$

Therefore,  $R(a\psi_1 + b\psi_2) = au_1 + bu_2$  it follows that  $R$  is linear. Since by Riesz Representation theorem there exists a unique  $u$  such that  $R\psi = u$  therefore  $R$  is obviously a bijection so it's an isometry since  $\|R\psi\| = \|\psi\|_*$  by Riesz Representation theorem. □

## 1.4 The Lax-Milgram Theorem

**Definition 1.19.** (Bilinear forms) A bilinear form  $a(\cdot, \cdot)$  on a vector space  $V$  is a real-valued function on the Cartesian product  $V \times V$  that is linear in both arguments, i.e.

$$a(u + \lambda w, v) = a(u, v) + \lambda a(w, v), \quad a(u, v + \lambda w) = a(u, v) + \lambda a(u, w)$$

for all  $u, v, w \in V$ , and any  $\lambda \in \mathbb{R}$ .

**Definition 1.20.** A bilinear form  $a(\cdot, \cdot)$  is said to be bounded if there exists a positive constant  $c_1 > 0$  such that

$$|a(w, v)| \leq c_1 \|w\| \|v\| \quad \forall w, v \in V$$

A bilinear form  $a(\cdot, \cdot)$  is said to be coercive on  $V$  if there exists a positive constant  $c_0 > 0$  such that

$$c_0 \|v\|^2 \leq a(v, v) \quad \forall v \in V$$

**Theorem 1.21.** Let  $V$  be a Hilbert space with inner product  $(\cdot, \cdot)$  and let  $a(\cdot, \cdot)$  be a bilinear form on  $V$  and is bounded and coercive. Then, for any bounded linear functional  $\psi \in V^*$ , there exists a unique  $u \in V$  that solves

$$a(u, v) = \psi(v) \quad \forall v \in V$$

In addition,

$$\|u\| \leq \frac{1}{c_0} \|\psi\|_*$$

**Proof.** (Proof by contraction mapping). Let  $A : V \rightarrow V^*$  be the operator defined by  $Aw(\cdot) = a(w, \cdot)$  for each  $w \in V$ . In other words, for a given  $w \in V$ , the linear functional  $Aw$  maps each  $v \in V$  to  $a(w, v)$ . Note that the boundness of  $A$  follows since

$$\|Aw\|_* = \sup_{v \in V, \|v\| \leq 1} |a(w, v)| \leq c_1 \|w\|$$

and thus the operator norm  $\|A\|_{\mathcal{L}(V, V^*)} \leq c_1$ . Since  $A : V \rightarrow V^*$  and the Riesz map  $R : V^* \rightarrow V$ , it follows that the composition  $RA$  is a bounded linear operator from  $V$  to  $V$  and furthermore

$$\|RAw\| = \|Aw\|_* \leq c_1 \|w\| \quad \forall w \in V$$

Furthermore, it follows from the definitions of  $R$  and  $A$  that

$$(RAw, v) = a(w, v) \quad \forall w, v \in V$$

Next, for a parameter  $\lambda > 0$  to be chosen later, we define the mapping  $T_\lambda : V \rightarrow V$  by  $T_\lambda(v) = v - \lambda R(Av - \psi)$ . Notice that  $T_\lambda(v)$  is indeed in  $V$  since  $R(Av - \psi) \in V$ . Furthermore,  $u \in V$  solves the equation if and only if

$$Au = \psi \Leftrightarrow R(Au - \psi) = 0 \Leftrightarrow T_\lambda(u) = u$$

so  $u \in V$  is a solution if and only if  $u$  is a fixed point of  $T_\lambda$ . We now aim to show that the parameter  $\lambda$  can be chosen so that  $T_\lambda$  becomes a contraction. For any  $w$  and  $v \in V$ , setting  $z = w - v$  then

$$\begin{aligned} \|T_\lambda(w) - T_\lambda(v)\|^2 &= \|z - \lambda RAz\|^2 = \|z\|^2 - 2\lambda(RAz, z) + \lambda^2 \|RAz\|^2 \\ &= \|z\|^2 - 2\lambda a(z, z) + \lambda^2 \|Az\|_*^2 \\ &\leq (1 - 2\lambda c_0 + \lambda^2 c_1^2) \|z\|^2 \end{aligned}$$

choosing  $\lambda = c_0/c_1^2$ , we get  $\|T_\lambda(w) - T_\lambda(v)\| \leq \rho \|w - v\|$  where  $\rho = \sqrt{1 - c_0^2/c_1^2} < 1$ . Hence,  $T_\lambda$  is a contraction, and the Banach Fixed Point Theorem shows that there exists a unique fixed point  $u \in V$  of  $T_\lambda$ , which is the unique solution of the equation. Finally, we choose  $v = u$  in the equation and use coercivity to get

$$\|u\|^2 \leq \frac{1}{c_0} a(u, u) = \frac{1}{c_0} \psi(u) \leq \frac{1}{c_0} \|\psi\|_* \|u\|$$

□

# 2 Lebesgue and Sobolev Spaces

## 2.1 Spaces of continuously differentiable functions

**Definition 2.1.** (Multi-index notation) A multi-index  $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ . The order of  $\alpha$  is  $|\alpha| = \alpha_1 + \dots + \alpha_n$ . The notation

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}$$

**Example.**  $D^{(0,1)}f = \frac{\partial f}{\partial x_1}$ ,  $D^{(1,1)}f = \frac{\partial^2 f}{\partial x_1 \partial x_2}$ .

**Definition 2.2.** Let  $\Omega \subset \mathbb{R}^n$ ,  $n \geq 1$  open. For each  $k \geq 0$ ,  $C^k(\Omega)$  is the space of functions  $v : \Omega \rightarrow \mathbb{R}$  such that  $D^\alpha v$  exists and it is continuous on  $\Omega$  for all  $|\alpha| \leq k$ .

**Definition 2.3.**  $C^k(\bar{\Omega})$  is the space of functions  $v : \Omega \rightarrow \mathbb{R}$  such that  $D^\alpha v$  exists, is bounded and uniformly continuous on  $\Omega$  for all  $|\alpha| \leq k$ .

**Remark.** The notation  $C^k(\bar{\Omega})$  comes from the fact that functions on  $C^k(\bar{\Omega})$  has a unique extension to  $\bar{\Omega}$ . However, if  $\Omega$  is unbounded set, this definition differs from the space of continuously differentiable functions defined over  $\bar{\Omega}$ . For example, if  $\Omega = \mathbb{R}^n$  then  $C^k(\bar{\Omega}) = C^k(\mathbb{R}^n)$  so we get  $C^k(\mathbb{R}^n) \neq C^k(\bar{\mathbb{R}^n})$ .

**Definition 2.4.** A norm on  $C^k(\bar{\Omega})$  can be defined by

$$\|v\|_{C^k(\bar{\Omega})} = \max_{|\alpha| \leq k} \sup_{x \in \Omega} |D^\alpha v(x)|$$

**Definition 2.5.** Spaces of smooth functions:  $C^\infty(\Omega) = \bigcap_{k \geq 0} C^k(\Omega)$ ,  $C^\infty(\bar{\Omega}) = \bigcap_{k \geq 0} C^k(\bar{\Omega})$ .

**Definition 2.6.** The support of  $v : \Omega \rightarrow \mathbb{R}$  is defined by

$$\text{supp}(v) = \overline{\{x \in \Omega : v(x) \neq 0\}}$$

We say that  $v$  is compactly supported in  $\Omega$  if  $\text{supp}(v)$  is compact subset of  $\Omega$ .

**Definition 2.7.** The space  $C_0^\infty(\Omega)$  is the space of functions in  $C^\infty(\Omega)$  that are compactly supported in  $\Omega$ .

**Exercise.** If  $v \in C_0^\infty(\Omega)$ , show that  $D^\alpha v \in C_0^\infty(\Omega)$  for all  $|\alpha| \leq k$ .

## 2.2 Lebesgue spaces

(Equation almost everywhere) Often, we will deal with functions that are not identically equal but that differ only on a set of measure zero:  $v, w : \Omega \rightarrow \mathbb{R}$ ,  $\{x \in \Omega : v(x) \neq w(x)\}$  is the set of measure zero. In this case, we say that they are equal almost everywhere in the domains. We saw previously the example of  $\|L^2[a, b]\|$  with norm  $\|v\|_{L^2[a, b]} = (\int_a^b |v(x)|^2 dx)^{1/2}$ . However,  $\|v\|_{L^2[a, b]} = 0$  for any function  $v$  that is equal to 0 almost everywhere. For example,  $(a, b) = (0, 1)$ ,

$$v(x) = \begin{cases} 0 & \text{if } x \neq \frac{1}{2} \\ \infty & \text{if } x = \frac{1}{2} \end{cases} \implies \int_a^b |v(x)|^2 dx = 0$$

Strictly speaking,  $\|\cdot\|_{L^2[a, b]}$  is not a norm functions on  $(a, b)$  so we avoid this difficulties by considering any two functions to be the same if and only if they are equal almost everywhere. We write  $v = w$  if and only if  $v$  and  $w$  almost everywhere. We can do this as long as everything we do with functions does not depend on values of sets of measure zero. We can take integrals:

$$\int_\Omega v(x) dx = \int_\Omega w(x) dx$$

**Definition 2.8.**  $L^p(\Omega)$  with  $p \in [1, \infty)$ . For each  $p \in [1, \infty)$ ,  $L^p(\Omega)$  is the space of functions:  $v : \Omega \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$  that are Lebesgue measurable and such  $\int_\Omega |v|^p dx < \infty$ .

**Definition 2.9.** (Essentially bounded functions) A Lebesgue measurable function  $v : \Omega \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$  is called essentially bounded if there exists a constant  $M \geq 0$  such that  $|v(x)| \leq M$  for almost every  $x \in \Omega$ . The infimum over all such constants  $M$  is called the essential supremum  $\text{ess sup}_{x \in \Omega} |v(x)|$ .

**Definition 2.10.**  $L^\infty(\Omega)$  is the space of Lebesgue measurable functions  $v : \Omega \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$  that are essentially bounded on  $\Omega$ .

**Definition 2.11.** We define the norms  $\|\cdot\|_{L^p(\Omega)}$  is defined by

$$\|v\|_{L^p(\Omega)} = \begin{cases} (\int_{\Omega} |v|^p dx)^{1/p} & \text{if } p \in [1, \infty) \\ \text{ess sup}_{x \in \Omega} |v(x)| & \text{if } p = \infty \end{cases}$$

The space  $L^p(\Omega)$  is a Banach space under its norm for all  $p \in [1, \infty]$ . For  $p = 2$ ,  $L^2(\Omega)$  is a Hilbert space with inner-product

$$(v, w)_{L^2(\Omega)} = \int_{\Omega} vw dx$$

for all  $v, w \in L^2(\Omega)$ .

**Theorem 2.12.** (Hölder inequality) Let  $p \in [1, \infty]$ , let  $q \in [1, \infty]$  such that  $\frac{1}{p} + \frac{1}{q} = 1$  then

$$\int_{\Omega} |vw| dx \leq \|v\|_{L^p(\Omega)} \|w\|_{L^q(\Omega)}$$

for all  $v \in L^p(\Omega)$  and  $w \in L^q(\Omega)$ .

**Proof.** See problem sheet 1. □

**Theorem 2.13.** (Minkowski) For  $p \in [1, \infty]$ , we have

$$\|v + w\|_{L^p(\Omega)} \leq \|v\|_{L^p(\Omega)} + \|w\|_{L^p(\Omega)}$$

for all  $v, w \in L^p(\Omega)$ .

**Proof.** For  $p < \infty$ , consider  $v, w \in L^p(\Omega)$  arbitrary, let  $q$  satisfies  $\frac{1}{p} + \frac{1}{q} = 1$  then  $q = \frac{p}{p-1}$ . Let  $z = (v + w)^{p-1}$  then  $|z|^q = |v + w|^p$  so

$$\begin{aligned} \int_{\Omega} |z|^p dx &= \int_{\Omega} |v + w|^p dx < \infty \\ \|v + w\|_{L^p(\Omega)}^p &= \int_{\Omega} |v + w|^p dx = \int_{\Omega} |v + w|^{p-1} |v + w| dx \\ &\leq \int_{\Omega} |v + w|^{p-1} (|v| + |w|) dx = \int_{\Omega} |v + w|^{p-1} |v| dx + \int_{\Omega} |v + w|^{p-1} |w| dx \\ &= \int_{\Omega} |zv| dx + \int_{\Omega} |zw| dx \leq \|z\|_{L^q(\Omega)} \|v\|_{L^p(\Omega)} + \|z\|_{L^q(\Omega)} \|w\|_{L^p(\Omega)} \\ &= \|z\|_{L^q(\Omega)} (\|v\|_{L^p(\Omega)} + \|w\|_{L^p(\Omega)}) = \|v + w\|_{L^p(\Omega)}^{p-1} (\|v\|_{L^p(\Omega)} + \|w\|_{L^p(\Omega)}) \end{aligned}$$

If  $v + w = 0$  then trivial. Now if  $v + w \neq 0$  then,

$$\|v + w\|_{L^p(\Omega)} \leq \|v\|_{L^p(\Omega)} + \|w\|_{L^p(\Omega)}$$

□

## 2.3 Sobolev spaces

(Weak derivatives) Consider a classically differentiable function  $v \in C^k(\Omega)$  with the partial derivative  $D^\alpha v$ , for some multi-index  $\alpha$  of order at most  $k$ . Multiplying  $D^\alpha v$  by a test function  $\phi \in C_0^\infty(\Omega)$  and integrating by parts  $|\alpha|$ -times leads to the identity

$$\int_{\Omega} D^\alpha v \phi dx = (-1)^{|\alpha|} \int_{\Omega} v D^\alpha \phi dx, \quad \forall \phi \in C_0^\infty(\Omega)$$

where all boundary terms vanish due to the compact support of  $\phi$  in  $\Omega$ .

**Definition 2.14.** Let  $\alpha$  be a multi-index and let  $v \in L^p(\Omega)$ ,  $p \in [1, \infty]$ . Suppose that there exists a function  $w \in L^1(\Omega)$  such that

$$\int_{\Omega} w \phi dx = (-1)^{|\alpha|} \int_{\Omega} v D^\alpha \phi dx, \quad \forall \phi \in C_0^\infty(\Omega)$$

Then we say that  $w$  is the weak derivative of  $v$  of index  $\alpha$  and we write  $w = D^\alpha v$ . The function  $v$  is said to be weakly differentiable to order  $k$  if the weak derivative  $D^\alpha v$  exists for each multi-index  $\alpha$  with  $|\alpha| \leq k$ .

**Remarks.** Properties of weak derivatives:

1. Weak derivative  $D^\alpha v$  is unique if it exists which can be proved by Du Bois Reymond Lemma below.
2. Weak derivatives are consistent with classical ones: if  $v \in C^k(\Omega)$  then its classical and weak derivatives coincide

a.e. in  $\Omega$ .

Heuristic explanation of how weak derivative is consistent with classical notation of derivative. Suppose  $v \in C^1(a, b)$ , consider  $x \in (a, b)$ . We can choose  $\phi = \phi_{x,h}$  such that

$$\phi_{x,h}(y) \approx \frac{1}{h} \max(1 - \frac{|x-y|}{h}, 0)$$

where  $h > 0$ . Also required  $\int_a^b \phi_{x,h} dy = 1, \forall h > 0$ . Then

$$\lim_{h \rightarrow 0} \int_a^b v'(y) \phi_{x,h}(y) dy = v'(x)$$

also have

$$\begin{aligned} \int_a^b v \phi'_{x,h}(y) dy &\approx \frac{1}{h^2} \int_{x-h}^x v dy - \frac{1}{h^2} \int_x^{x+h} v dy \\ &\approx \frac{1}{h} \left( \frac{1}{h} \int_{x-h}^x v dy - \frac{1}{h} \int_x^{x+h} v dy \right) \end{aligned}$$

where

$$v'(x) \approx \lim_{h \rightarrow 0} \frac{\frac{1}{h} \int_x^{x+h} v dy - \frac{1}{h} \int_{x-h}^x v dy}{h} \approx \lim_{h \rightarrow 0} \frac{v(x+h) - v(x)}{h}$$

**Definition 2.15.** We define the locally integrable functions on an open subset  $\Omega$  of  $\mathbb{R}^n$  by

$$L^1_{\text{loc}} := \{f : \Omega \rightarrow \mathbb{R}; \forall x \in \Omega, \exists r > 0 \text{ s.t. } B(x, r) \subseteq \Omega, f|_{B(x,r)} \in L^1(B(x, r))\}$$

**Lemma 2.16.** Let  $u \in L^1_{\text{loc}}(\mathbb{R}^n)$ ,  $\rho \in C_0^\infty(\mathbb{R}^n)$ . We define the convolution of  $u$  and  $\rho$  by

$$\rho * u(x) := \int_{\mathbb{R}^n} \rho(x-y) u(y) dy = \int_{\mathbb{R}^n} \rho(y) u(x-y) dy$$

then  $\rho * u \in C^\infty(\mathbb{R}^n)$  and for all  $\mathbb{N}_0^n$  one has

$$D^\alpha(\rho * u) = (D^\alpha \rho) * u$$

**Proof.** First, notice that the integral is well-defined since  $\rho$  is bounded and has compact support. To show the continuity of  $\rho * u$ , there exists some  $R > 0$  such that  $\text{supp} \rho \subseteq B(0, R)$ . Let  $R' > 0, \delta > 0$ . For  $x, x' \in B(0, R'), |x-x'| < \delta$ , one obtains

$$\begin{aligned} |\rho * u(x) - \rho * u(x')| &= \left| \int_{B(0, R+R')} (\rho(x-y) - \rho(x'-y)) u(y) dy \right| \\ &\leq \sup\{|\rho(z) - \rho(z')|; |z - z'| < \delta\} \int_{B(0, R+R')} u(y) dy \end{aligned}$$

since the second factor is finite since  $u$  is locally integrable and the first factor becomes small when  $\delta$  is small since  $\rho$  is uniformly continuous. The equality is as a result of the differentiable under the integral within different parameters.  $\square$

**Definition 2.17.** A sequence  $(\rho_k)_{k \in \mathbb{N}}$  in  $C_0(\mathbb{R}^n)$  is called a  $\delta$ -sequence if  $\rho_k \geq 0, \int \rho_k(x) dx = 1$  and  $\text{supp} \rho_k \subseteq B[0, 1/k]$  for all  $k \in \mathbb{N}$ .

**Lemma 2.18** (Not covered in the lecture, my own one). (Du Bois Reymond Lemma) For any two functions  $w$  and  $\tilde{w} \in L^1(\Omega)$ ,

$$\int_\Omega (\tilde{w} - w) \phi dx = 0 \quad \forall \phi \in C_0^\infty(\Omega) \Leftrightarrow w = \tilde{w} \text{ a.e. in } \Omega$$

**Proof.** It suffices to show that if  $f \in L^1_{\text{loc}}(a, b)$  and

$$\int_{(a,b)} f \phi' dx = 0 \quad \forall \phi \in C_0^\infty(a, b)$$

then  $f \equiv c$  (constant) almost everywhere. Let  $x$  and  $\tilde{x}$  be Lebesgue points for  $f \in [f_0]$ , assume  $a < x < \tilde{x} < b$ , take us take  $\delta > 0$  small and fixed so that

$$\bar{\phi}(\xi) = 2\chi_I(\xi) + [1 + \frac{1}{\delta}(\xi - x)]\chi_T(\xi) - [1 - \frac{1}{\delta}(\xi - \tilde{x})]\chi_{\tilde{T}}(\xi)$$

with  $I = (x + \delta, \tilde{x} - \delta)$ ,  $T = (x - \delta, x + \delta)$  and  $\tilde{T} = (\tilde{x} - \delta, \tilde{x} + \delta)$ . Then define  $\mu_\varepsilon \in C_0^\infty(-\varepsilon, \varepsilon)$  satisfies  $\int_{\mathbb{R}} \mu dx = 1$  with  $\varepsilon < \delta$  then  $\mu_\varepsilon * \bar{\phi} \in C_0^\infty(a, b)$  with

$$\mu_\varepsilon * \bar{\phi} = \begin{cases} 0, & \xi \in (a, b) \setminus (x - \delta - \varepsilon, \tilde{x} + \delta + \varepsilon) \\ 2, & \xi \in [x + \delta + \varepsilon, \tilde{x} - \delta - \varepsilon] \end{cases}$$

Also,

$$\frac{d}{dx}(\mu_\varepsilon * \bar{\phi}) = \mu_\varepsilon \bar{\phi}' = 0$$

on the interiors

$$(a, b) \setminus [x - \delta - \varepsilon, \tilde{x} + \delta + \varepsilon] \text{ and } (x + \delta + \varepsilon, \tilde{x} - \delta - \varepsilon)$$

with

$$-\frac{1}{\delta} \leq \frac{d}{dx}(\mu_\varepsilon * \bar{\phi}) \leq \frac{1}{\delta} \quad \forall x \in (a, b)$$

Let us see what happens in the portions

$$(x - \delta + \varepsilon, x + \delta - \varepsilon) \text{ and } (\tilde{x} - \delta + \varepsilon, \tilde{x} + \delta - \varepsilon)$$

For  $\xi$  in the interval around  $x$  we have

$$\begin{aligned} \mu_\varepsilon * \bar{\phi}(\xi) &= \int_t \mu_\varepsilon(t) \bar{\phi}(\xi - t) dt \\ &= \int_t \mu_\varepsilon(t) [1 + \frac{1}{\delta}(\xi - t - x)] dx \\ &= 1 + \frac{1}{\delta}(\xi - x) - \frac{1}{\delta} \int_t t \mu_\varepsilon(t) dt \\ &= \bar{\phi}(\xi) - m(\varepsilon) \end{aligned}$$

where

$$m(\varepsilon) = \frac{1}{\delta} \int_t t \mu_\varepsilon(t) dt = \frac{1}{\delta} \int_{-\varepsilon}^\varepsilon t \mu_\varepsilon(t) dt \text{ satisfies } |m(\varepsilon)| < \frac{\varepsilon}{\delta}$$

Similarly, for  $\xi$  around  $\tilde{x}$  we have  $\mu_\varepsilon * \bar{\phi} = \bar{\phi} + m(\varepsilon)$ . We want to show that

$$\int f(\mu_\varepsilon * \bar{\phi})' dx = 0$$

and take the limit to yield

$$\int f \bar{\phi}' dx = 0$$

Let us estimate

$$\begin{aligned} \left| \int f(\mu_\varepsilon * \bar{\phi})' - \int f \bar{\phi}' \right| &= \int |f[(\mu_\varepsilon * \bar{\phi})' - \bar{\phi}']| dx \\ &\leq \int |f| |(\mu_\varepsilon * \bar{\phi})' - \bar{\phi}'| dx \\ &= \int_{(x-\delta-\varepsilon, \tilde{x}+\delta+\varepsilon)} |f| |(\mu_\varepsilon * \bar{\phi})' - \bar{\phi}'| dx \end{aligned}$$

In addition, we know that for  $\varepsilon$  small the interval  $J = (x - \delta - \varepsilon, \tilde{x} + \delta + \varepsilon) \subset (a, b)$  so  $f \in L^1(J)$ . Furthermore,  $|\mu_\varepsilon * \bar{\phi})' - \bar{\phi}'|$  is bounded for all of  $(a, b)$  and satisfies

$$\lim_{\varepsilon \rightarrow 0} |(\mu_\varepsilon * \bar{\phi})' - \bar{\phi}'| = 0$$

for every  $\xi \in (a, v)$ . By dominated convergence theorem,

$$\lim_{\varepsilon \rightarrow 0} \int_{(x-\delta-\varepsilon, \tilde{x}+\delta+\varepsilon)} |f| |(\mu_\varepsilon * \bar{\phi})' - \bar{\phi}'| dx = \int_{(x-\delta-\varepsilon, \tilde{x}+\delta+\varepsilon)} \lim_{\varepsilon \rightarrow 0} |f| |(\mu_\varepsilon * \bar{\phi})' - \bar{\phi}'| dx = 0$$

then we have

$$\int f \bar{\phi}' dx = 0$$

In addition,

$$\bar{\phi}'(\xi) = \frac{1}{\delta} \chi_T(\xi) - \frac{1}{\delta} \chi_{\tilde{T}}(\xi)$$

almost everywhere except to the four corners. Substitute into the previous equation to derive

$$\frac{1}{\delta} \int_{(x-\delta, x+\delta)} f - \frac{1}{\delta} \int_{(\tilde{x}-\delta, \tilde{x}+\delta)} f = 0$$

take  $\delta \rightarrow 0$  we have  $f(x) = f(\tilde{x})$  almost everywhere which is the result required.  $\square$

**Definition 2.19.** (Sobolev spaces) Let  $k \geq 0$  be an integer and  $p \in [1, \infty]$ . The space  $W^{k,p}(\Omega)$  is the space of functions in  $L^p(\Omega)$  that are weakly differentiable up to order  $k$  such that the weak derivative  $D^\alpha v \in L^p(\Omega)$  for all  $|\alpha| \leq k$ . We write

$$W^{k,p}(\Omega) = \{v \in L^p(\Omega), D^\alpha v \in L^p(\Omega), \forall |\alpha| \leq k\}$$

$W^{k,p}(\Omega)$  is called a Sobolev space of order  $k$ . For the special case  $k = 0$ , this reduces simply to  $L^p(\Omega)$ . We define the norms

$$\|v\|_{W^{k,p}(\Omega)} = \begin{cases} (\sum_{|\alpha| \leq k} \|D^\alpha v\|_{L^p(\Omega)}^p)^{\frac{1}{p}}, & p \in [1, \infty) \\ \max_{|\alpha| \leq k} \|D^\alpha v\|_{L^\infty(\Omega)}, & p = \infty. \end{cases}$$

Furthermore, we define the semi-norms

$$|v|_{W^{k,p}(\Omega)} = \begin{cases} (\sum_{|\alpha|=k} \|D^\alpha v\|_{L^p(\Omega)}^p)^{\frac{1}{p}}, & p \in [1, \infty) \\ \max_{|\alpha|=k} \|D^\alpha v\|_{L^\infty(\Omega)}, & p = \infty. \end{cases}$$

Similar to  $L^p(\Omega)$ , the Sobolev space  $W^{k,p}(\Omega)$  is a Banach space under the norm  $\|\cdot\|_{W^{k,p}(\Omega)}$  for all  $k \geq 0$  and  $p \in [1, \infty]$ .

**Example.**  $v(x) = \max(1 - |x|, 0)$ ,  $w \in W^{k,p}(\Omega)$ . For  $k = 0$ ,  $v \in L^p(\Omega)$ ,  $\forall p \in [1, \infty]$  since  $|v(x)| \leq 1$ ,  $\forall x \in \Omega$ . For  $k = 1$ ,

$$v' = \begin{cases} 1 & x \in (-1, 0) \\ -1 & x \in (0, 1) \\ 0 & \text{otherwise} \end{cases} \\ \implies |v'(x)| \leq 1 \text{ a.e. in } \Omega$$

thus,  $v' \in L^p(\Omega)$ ,  $\forall p \in [1, \infty]$ . However the weak derivative of  $v'$  does not exist. Let  $\phi \in C_0^\infty(\Omega)$  then

$$\begin{aligned} \int_{\Omega} v' \phi' dx &= \int_{-1}^0 \phi'(x) dx - \int_0^1 \phi'(x) dx \\ &= (\phi(0) - \phi(-1)) - (\phi(1) - \phi(0)) \\ &= 2\phi(0) - \phi(-1) - \phi(1) \end{aligned}$$

If  $v'$  were weakly differentiable,  $\exists w$  such that

$$\int_{\Omega} w \phi dx = 2\phi(0) - \phi(-1) - \phi(1)$$

However, this leads to a contradiction for suitable choice of  $\phi$  for example take  $\phi(0) \rightarrow \infty$  but  $\int_{\Omega} w \phi dx$  is finite.

$$\implies v \in W^{1,p}(\Omega), \quad \forall p \in [1, \infty]$$

but  $v \notin W^{2,p}(\Omega)$  or higher since  $v'$  is not weakly differentiable.

**Example.** Consider  $v(x) = x^{1/2}$  and  $\Omega = (0, 1)$ . For  $k = 0$ ,  $v(x) = x^{1/2} \in L^p(\Omega)$ ,  $\forall p \in [1, \infty]$  since  $|v(x)| \leq 1$  on  $(0, 1)$ . For  $k = 1$ ,  $v' = \frac{1}{2}x^{-1/2}$ .  $v' \notin L^\infty(\Omega)$  and

$$\int_0^1 |v'(x)|^p dx = \int_0^1 x^{-p/2} dx$$

is finite if and only if  $p < 2$ . Thus,  $v(x) \in W^{1,p}$  for  $p \in [1, 2)$ . For  $k = 2$ ,  $v''(x) = -\frac{1}{4}x^{-3/2}$  then  $v''(x) \notin L^p(\Omega)$  for  $p \in [1, \infty]$ .

**Density of smooth functions:** In the following, we restrict our attention to some domains  $\Omega$  that have sufficiently regular boundaries. If  $\Omega$  is sufficiently regular, then functions in  $C^\infty(\bar{\Omega})$  are dense in  $W^{k,p}(\Omega)$  for all value of  $k > 0$  and  $p \in [0, \infty)$ . i.e., for any  $v \in W^{k,p}(\Omega)$ , there exists  $(v_j) \subset C^\infty(\bar{\Omega})$  and  $\|v - v_j\|_{W^{k,p}(\Omega)} \rightarrow 0$  as  $j \rightarrow \infty$ . The case  $p = \infty$  is excluded since if we take

$$v(x) = \begin{cases} 0 & \text{if } x \in (-1, 0) \\ 1 & \text{if } x \in (0, 1) \end{cases}$$

there does not exist  $(v_j) \subset C^\infty[-1, 1]$  such that  $v_j \rightarrow v$  in  $L^\infty(-1, 1)$  since uniform limit of continuous function on  $C^\infty[-1, 1]$  is continuous, we would deduce  $v$  must also be continuous which is a contradiction. The main consequence of density for our purposes is that we can prove results about  $W^{k,p}(\Omega)$  by first showing it for smooth functions, and then extending to general functions in  $W^{k,p}(\Omega)$ .

**Theorem 2.20.** Let  $(a, b)$  be a bounded interval, let  $p \in [1, \infty]$ . There exists a constant  $C$  depending only on the length of the interval  $(a, b)$  and  $p$  such that

$$\|v\|_{L^\infty(a,b)} \leq C \|v\|_{W^{1,p}(a,b)}, \quad \forall v \in W^{1,p}(a,b)$$

**Proof.** For  $p = \infty$  this is immediate by definition. For  $p < \infty$ , there are several steps:

1. Prove it for smooth functions. Consider  $v \in C^\infty[a, b]$ , since  $v$  is smooth then

$$\|v\|_{L^\infty[a,b]} = \max_{x \in [a,b]} |v(x)|$$

Let  $x \in [a, b]$  be arbitrary, use the mean value theorem,  $\exists y \in [a, b]$  such that

$$v(y) = \frac{1}{b-a} \int_a^b v(s) ds$$

and use the fundamental theorem of calculus,

$$v(x) = v(y) + \int_y^x v'(s) ds$$

$$\implies |v(x)| \leq |v(y)| + \int_y^x |v'(s)| ds \leq |v(y)| + \int_a^b |v'(s)| ds$$

$$\begin{aligned} &\Rightarrow |v(x)| \leq \frac{1}{b-a} \int_a^b |v(s)| ds + \int_a^b |v'(s)| ds \\ &\leq \max(1, \frac{1}{b-a}) (\int_a^b |v(s)| ds + \int_a^b |v'(s)| ds) = \max(1, \frac{1}{b-a}) \|v\|_{W^{1,1}(a,b)} \end{aligned}$$

By Hölder inequality,

$$\begin{aligned} |v(x)| &\leq \max(1, \frac{1}{b-a}) (\|1\|_{L^q(a,b)} \|v\|_{L^p(a,b)} + \|1\|_{L^q(a,b)} \|v'\|_{L^p(a,b)}) \\ &= \max(1, \frac{1}{b-a}) (b-a)^{1-\frac{1}{p}} (\|v\|_{L^p(a,b)} + \|v'\|_{L^p(a,b)}) \end{aligned}$$

in this case,

$$|v(x)| \leq 2 \max(\frac{1}{b-a}, 1) (b-a)^{1-\frac{1}{p}} \|v\|_{W^{1,p}(a,b)}, \quad \forall v \in C^\infty[a, b]$$

Since the RHS of the inequality is independent of  $x \in [a, b]$ ,

$$\Rightarrow \|v\|_{L^\infty(a,b)} = \max_{x \in [a,b]} |v(x)| \leq 2 \max(\frac{1}{b-a}, 1) (b-a)^{1-\frac{1}{p}} \|v\|_{W^{1,p}(a,b)}, \quad \forall v \in C^\infty[a, b]$$

2.  $\|v\|_{L^\infty(a,b)} = \lim_{j \rightarrow \infty} \|v_j\|_{L^\infty(a,b)} \leq 2 \max(\frac{1}{b-a}, 1) (b-a)^{1-\frac{1}{p}} \|v_j\|_{W^{1,p}(a,b)}$  Since  $C^\infty[a, b]$  are dense in  $W^{1,p}(a, b)$  and since  $C[a, b]$  is closed in  $L^\infty(a, b)$ , we can show that any  $v \in W^{1,p}(a, b)$  is equal almost everywhere in  $(a, b)$  to a continuous function  $[a, b]$ .

□

### Remarks.

- (a) Functions with discontinuity that cannot be removed by changing values on sets of zero measure are not in  $W^{1,p}(a, b)$ .
- (b) From now on, we can assign a notion of point value to functions  $W^{1,p}(a, b)$ : given  $v \in W^{1,p}(a, b)$ , let's define its point value  $v(x)$  where  $x \in [a, b]$  as the values given by its continuous representatives.

**Definition 2.21.** (Hilbert spaces  $H^k(\Omega)$ )  $H^k(\Omega) = W^{k,2}(\Omega)$ , where the norm is defined by

$$\|\cdot\|_{H^k(\Omega)} = \|\cdot\|_{W^{k,2}(\Omega)}$$

and this is a Hilbert space with inner product given by

$$(u, v)_{H^k(\Omega)} = \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}, \quad \forall u, v \in H^k(\Omega)$$

For  $k = 1$ , we also a convenient notation: let  $\nabla v$  denote the vector of first weak derivatives of  $v$

$$\nabla v = (D^{(1,0,\dots,0)}, \dots, D^{(0,0,\dots,1)})$$

and we define

$$\|\nabla v\|_{L^2(\Omega)} := \| |\nabla v| \|_{L^2(\Omega)} = \left( \int_{\Omega} |\nabla v|^2 dx \right)^{1/2} = \|v\|_{H^1(\Omega)}$$

then

$$\|v\|_{H^1(\Omega)} = (\|v\|_{L^2(\Omega)}^2 + \|v\|_{H^1(\Omega)}^2)^{1/2} = (\|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2)^{1/2}$$

In the analysis of PDE, we will often work with subspace of  $H^k(\Omega)$  that consist of functions satisfying some boundary conditions a simple example is as following

**Definition 2.22.** (The space  $H_0^1(\Omega)$ ) Define  $H_0^1(\Omega)$  as the closure of  $C_0^\infty(\Omega)$  in the norm of  $H^1(\Omega)$ . For the special case of  $\Omega = (a, b)$  we have

$$H_0^1(a, b) = \{v \in H^1(a, b) : v(a) = v(b) = 0\}$$

**Example.**  $v = \max(1 - |x|, 0)$  on  $\Omega = (-2, 2)$  then  $v \in H_0^1(-2, 2)$ .

**Theorem 2.23.** Let  $\Omega$  be a bounded open set in  $\mathbb{R}^n$ . Then  $\exists$  a constant  $C_p$  depends only on  $\Omega$ , such that

$$\|v\|_{L^2(\Omega)} \leq C_p \|\nabla v\|_{L^2(\Omega)}, \quad \forall v \in H_0^1(\Omega)$$

**Proof.** Say  $\Omega = (a, b)$ . Here are steps to do so

- (1) Show the result for smooth functions on  $C_0^\infty(\Omega)$ . Let  $v \in C_0^\infty(\Omega)$ , then  $v(a) = 0$ . Use the result  $v(x) = \int_a^x v' ds$

$$\Rightarrow \|v\|_{L^2(a,b)}^2 = \int_a^b |v(x)|^2 dx = \int_a^b \left| \int_a^x v' ds \right|^2 dx$$

By Cauchy-Schwarz inequality

$$\left| \int_a^x v' ds \right| \leq \left( \int_a^x 1 ds \right)^{1/2} \left( \int_a^x |v'|^2 ds \right)^{1/2} \leq (x-a)^{1/2} \left( \int_a^x |v'|^2 ds \right)^{1/2}$$

$$\begin{aligned}
\Rightarrow \|v\|_{L^2(a,b)}^2 &\leq \int_a^b (x-a) \left( \int_a^b |v'|^2 ds \right) dx = \left( \int_a^b |v'|^2 ds \right) \left( \int_a^b (x-a) ds \right) = \frac{(b-a)^2}{2} \|v'\|_{L^2(a,b)}^2 \\
&\Rightarrow \|v\|_{L^2(a,b)} \leq \frac{b-a}{\sqrt{2}} \|v'\|_{L^2(a,b)} \\
&\Rightarrow C_p = \frac{b-a}{\sqrt{2}}, v \in C_0^\infty(a,b)
\end{aligned}$$

which complete the proof for  $\Omega = (a, b)$ .

□

# 3 Weak formulations of elliptic PDE

## 3.1 Weak formulations of second-order PDE

We are looking for PDE of the form

$$-\sum_{i,j=1}^n \frac{\partial}{\partial x_i} (a_{ij} \frac{\partial u}{\partial x_j}) + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i} + cu = f \text{ in } \Omega \subset \mathbb{R}^n$$

the functions  $a_{ij}, b_i, c, f$  are given data and  $u$  is unknown solution. In many cases, there is no classical solution  $u$  : e.g.  $u \notin C^2(\bar{\Omega})$  if some of the data is not continuous. This type of equation is called elliptic if

(a)  $a_{ij}(x) = a_{ji}(x) \forall x \in \Omega, i, j \in [1, n]$ .

(b)

$$\sum_{i,j=1}^n a_{ij}(x) v_i v_j \geq c_* \sum_{i=1}^n |v_i|^2 \quad \forall (v_1, \dots, v_n) \in \mathbb{R}^n, \forall x \in \Omega$$

where  $c_* > 0$ .

Equivalently the matrix

$$\begin{pmatrix} a_{11}(x) & \cdots & a_{1n}(x) \\ \vdots & \cdots & \vdots \\ a_{n1}(x) & \cdots & a_{nn}(x) \end{pmatrix}$$

is symmetric and also positive definite uniformly in  $\Omega$ . We start by considering this type of equation with the boundary condition  $u = 0$  on  $\partial\Omega$  (Homogeneous Dirichlet B.C.) To motivate the weak form of the solution, suppose  $u$  is a classical solution then multiply the equation by a test function  $\phi \in C_0^\infty(\Omega)$  and integrate on  $\Omega$  then

$$\int_{\Omega} (-\sum_{i,j=1}^n \frac{\partial}{\partial x_i} (a_{ij} \frac{\partial u}{\partial x_j})) \phi \, dx = \sum_{i,j=1}^n \int_{\Omega} a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial \phi}{\partial x_i} \, dx$$

then the equation becomes

$$\sum_{i,j=1}^n \int_{\Omega} a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial \phi}{\partial x_i} \, dx + \sum_{i=1}^n \int_{\Omega} b_i \frac{\partial u}{\partial x_i} \phi \, dx + \int_{\Omega} cu \phi \, dx = \int_{\Omega} f \phi \, dx$$

Moreover, if  $a_{ij}, b_i, c \in L^\infty(\Omega)$  and  $f \in L^2(\Omega)$  then we can extend above to all test functions  $v \in H_0^1(\Omega)$ .

Since we seek  $u = 0$  on  $\partial\Omega$ , we see that  $u \in H_0^1(\Omega)$  so we arrive at seek  $u \in H_0^1(\Omega)$  such that

$$\sum_{i,j=1}^n \int_{\Omega} a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial v}{\partial x_i} \, dx + \sum_{i=1}^n \int_{\Omega} b_i \frac{\partial u}{\partial x_i} v \, dx + \int_{\Omega} cuv \, dx = \int_{\Omega} f v \, dx, \quad \forall v \in H_0^1(\Omega) \quad (*)$$

**Definition 3.1.** Let  $a_{ij}, b_i, c \in L^\infty(\Omega) \forall 1 \leq i, j \leq n$  and  $f \in L^2(\Omega)$ . The weak formulation of the above equation is to find  $u \in H_0^1(\Omega)$  that solves  $(*) \forall v \in H_0^1(\Omega)$ . A solution  $u$  (if it exists) is called a weak solution of the original equation.

Application of the LMT: In order to apply the LMT, we require assumptions:

(a) Elliptic.  $a_{ij}, b_i, c \in L^\infty(\Omega) \forall 1 \leq i, j \leq n$  and  $f \in L^2(\Omega)$  where  $\Omega$  is open and bounded.

(b)  $b_i \in W^{1,\infty}(\Omega) \forall i \in [1, n]$  and  $c - \frac{1}{2} \sum_{i=1}^n \frac{\partial b_i}{\partial x_i} \geq 0$ .

Boundedness of  $a(\cdot, \cdot)$ , we want to show  $|a(w, v)| \leq c_1 \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \forall w, v \in H_0^1(\Omega)$ . By triangle inequality and let  $w, v \in H_0^1(\Omega)$  yields

$$|a(w, v)| \leq \sum_{i,j} \left| \int_{\Omega} a_{ij} \frac{\partial w}{\partial x_j} \frac{\partial v}{\partial x_i} \, dx \right| + \sum_i \left| \int_{\Omega} b_i \frac{\partial w}{\partial x_i} v \, dx \right| + \left| \int_{\Omega} cuv \, dx \right|$$

By Hölder inequality

$$\begin{aligned} \left| \int_{\Omega} a_{ij} \frac{\partial w}{\partial x_j} \frac{\partial v}{\partial x_i} \, dx \right| &\leq \int_{\Omega} |a_{ij}| \left| \frac{\partial w}{\partial x_j} \right| \left| \frac{\partial v}{\partial x_i} \right| \, dx \\ &\leq \|a_{ij}\|_{L^\infty(\Omega)} \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2(\Omega)} \\ &\leq \|a_{ij}\|_{L^\infty(\Omega)} \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2(\Omega)} \end{aligned}$$

Thus,

$$\begin{aligned}
\sum_{i,j} \left| \int_{\Omega} a_{ij} \frac{\partial w}{\partial x_j} \frac{\partial v}{\partial x_i} dx \right| &\leq \sum_{i,j} \|a_{ij}\|_{L^{\infty}(\Omega)} \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2(\Omega)} \\
&\leq \max_{i,j} \|a_{ij}\|_{L^{\infty}(\Omega)} \sum_{i,j} \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2(\Omega)} \\
&= C_a \left( \sum_i \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2(\Omega)} \right) \left( \sum_j \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \right)
\end{aligned}$$

where  $C_a = \max_{i,j} \|a_{ij}\|_{L^{\infty}(\Omega)}$ . Note

$$\sum_{j=1}^n \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \leq \sqrt{n} \left( \sum_j \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)}^2 \right)^{1/2} = \sqrt{n} \|\nabla w\|_{L^2(\Omega)}$$

then

$$C_a \left( \sum_i \left\| \frac{\partial v}{\partial x_i} \right\|_{L^2(\Omega)} \right) \left( \sum_j \left\| \frac{\partial w}{\partial x_j} \right\|_{L^2(\Omega)} \right) \leq C_a \|\nabla w\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}$$

since  $\|\nabla v\|_{L^2(\Omega)} \leq \|w\|_{H^1(\Omega)}$  then

$$\sum_{i,j} \left| \int_{\Omega} a_{ij} \frac{\partial w}{\partial x_j} \frac{\partial v}{\partial x_i} dx \right| \leq n C_a \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}$$

For the first order term, by Hölder inequality

$$\sum_i \left| \int_{\Omega} b_i \frac{\partial w}{\partial x_i} v dx \right| \leq \sum_i \|b_i\|_{L^{\infty}(\Omega)} \int_{\Omega} \left| \frac{\partial w}{\partial x_i} v \right| dx$$

By Cauchy-schwarz inequality,

$$\begin{aligned}
\sum_i \|b_i\|_{L^{\infty}(\Omega)} \int_{\Omega} \left| \frac{\partial w}{\partial x_i} v \right| dx &\leq \sum_i \|b_i\|_{L^{\infty}(\Omega)} \left\| \frac{\partial w}{\partial x_i} \right\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \\
&\leq \left( \sum_i \|b_i\|_{L^2(\Omega)}^2 \right)^{1/2} \left( \sum_i \left\| \frac{\partial w}{\partial x_i} \right\|_{L^2(\Omega)}^2 \right)^{1/2} \|v\|_{L^2(\Omega)} \\
&= \tilde{C}_b \|\nabla w\|_{L^2(\Omega)}^{1/2} \|v\|_{L^2(\Omega)} \\
&\leq \tilde{C}_b \|\nabla w\|_{H^1(\Omega)}^{1/2} \|v\|_{H^1(\Omega)}
\end{aligned}$$

Lastly,

$$\int_{\Omega} |c w v| dx \leq \|c\|_{L^{\infty}(\Omega)} \|w\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}$$

thus,

$$|a(w, v)| \leq (nC_a + \tilde{C}_b + \|c\|_{L^{\infty}(\Omega)}) \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \quad \forall w, v \in H_0^1(\Omega)$$

For the coercivity. Let  $v \in H_0^1(\Omega)$  to show  $a(v, v) \geq c_0 \|v\|_{H^1(\Omega)}^2$  note that

$$a(v, v) = \sum_{i,j} \int_{\Omega} a_{ij} \frac{\partial v}{\partial x_j} \frac{\partial v}{\partial x_i} dx + \sum_i \int_{\Omega} b_i \frac{\partial v}{\partial x_i} v dx + \int_{\Omega} c |v|^2 dx$$

then by ellipticity:

$$\begin{aligned}
\sum_{i,j} a_{ij}(x) p_i p_j &\geq c_* \sum_i |p_i|^2 \quad \forall (p_1, \dots, p_n) \in \mathbb{R}^n \\
\Rightarrow \sum_{i,j} \int_{\Omega} a_{ij} \frac{\partial v}{\partial x_j} \frac{\partial v}{\partial x_i} dx &\geq c_* \int_{\Omega} \sum_i \left| \frac{\partial v}{\partial x_i} \right|^2 dx = c_* \|\nabla v\|_{L^2(\Omega)}^2
\end{aligned}$$

In addition,

$$\begin{aligned}
\sum_i \int_{\Omega} b_i \frac{\partial v}{\partial x_i} v dx &= \sum_i \int_{\Omega} b_i \frac{1}{2} \frac{\partial}{\partial x_i} (|v|^2) dx = -\frac{1}{2} \sum_i \int_{\Omega} \frac{\partial b_i}{\partial x_i} |v|^2 dx \\
\Rightarrow \int_{\Omega} \sum_i b_i \frac{\partial v}{\partial x_i} v + c |v|^2 dx &= \int_{\Omega} \left( c - \frac{1}{2} \sum_i \frac{\partial b_i}{\partial x_i} \right) |v|^2 dx \geq 0 \quad \forall v \\
\Rightarrow a(v, v) &\geq C_* \|\nabla v\|_{L^2(\Omega)}^2
\end{aligned}$$

Recall Pointcaré inequality  $\|v\|_{L^2(\Omega)}^2 \leq C_p^2 \|\nabla v\|_{L^2(\Omega)}^2 \quad \forall v \in H_0^1(\Omega)$  which means

$$\begin{aligned}
\|v\|_{H^1(\Omega)}^2 &= \|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2 = (1 + C_p^2) \|\nabla v\|_{L^2(\Omega)}^2 \\
\Rightarrow a(v, v) &\geq \frac{C_*}{1 + C_p^2} \|v\|_{H^1(\Omega)}^2 \quad \forall v \in H_0^1(\Omega)
\end{aligned}$$

Now to check the boundedness of  $\ell(v)$

$$\begin{aligned}
|\ell(v)| &= \left| \int_{\Omega} f v dx \right| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \\
&\leq \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)} \quad \forall v \in H_0^1(\Omega)
\end{aligned}$$

**Theorem 3.2.** Let  $\Omega$  be a bounded open set in  $\mathbb{R}^n$ , let  $a_{ij} \in L^\infty(\Omega)$ ,  $b_i \in W^{1,\infty}(\Omega)$ ,  $c \in L^\infty(\Omega)$  and  $f \in L^2(\Omega)$  for all  $i, j \in [1, n]$ . Assume the  $a_{ij}$  satisfy the ellipticity condition that

$$c - \frac{1}{2} \sum_{i=1}^n \frac{\partial b_i}{\partial x_i} \geq 0 \text{ in } \Omega$$

Then, there exists a unique weak solution  $u \in H_0^1(\Omega)$  of (\*).

## 3.2 More general boundary conditions

There are several commonly encountered types of boundary condition on  $\partial\Omega$ .

1. (Inhomogeneous Dirichlet boundary condition)  $u = g(x)$  on  $\partial\Omega$  where  $g(x)$  is a prescribed function.
2. (Neumann boundary conditions)

$$\sum_{i,j=1}^n \nu_i a_{ij} \frac{\partial u}{\partial x_j} = g \text{ on } \partial\Omega$$

with  $\nu = (\nu_1, \dots, \nu_n)$  the outward unit normal on  $\partial\Omega$ , and where  $\alpha(x) > 0$  and  $g(x)$  are prescribed function.

3. (Robin boundary condition)

$$\sum_{i,j=1}^n \nu_i a_{ij} \frac{\partial u}{\partial x_j} + \alpha(x)u = g \text{ on } \partial\Omega,$$

with  $\nu = (\nu_1, \dots, \nu_n)$  the outward unit normal on  $\partial\Omega$ , and where  $\alpha(x) = 0$  and  $g(x)$  are prescribed function.

### 3.2.1 Inhomogeneous Dirichlet boundary condition

The case of the inhomogeneous condition  $u = g(x)$  on  $\partial\Omega$  can usually be reduced to the homogeneous case. Suppose that there exists a function  $\tilde{u} \in H^1(\Omega)$  such that  $\tilde{u} = g$  on  $\partial\Omega$ . Then, we write the solution

$$u = u_0 + \tilde{u}$$

where  $u_0 \in H_0^1(\Omega)$  is the solution of the homogeneous problem

$$a(u_0, v) = \ell(v) - a(\tilde{u}, v) \quad \forall v \in H_0^1(\Omega)$$

where

$$a(w, v) = \sum_{i,j} \int_{\Omega} a_{ij} \frac{\partial w}{\partial x_j} \frac{\partial v}{\partial x_i} dx + \sum_{i=1}^n \int_{\Omega} b_i \frac{\partial w}{\partial x_i} v dx + \int_{\Omega} c w v dx$$

and

$$\ell(v) = \int_{\Omega} f v dx$$

The existence of a unique solution  $u_0 \in H_0^1(\Omega)$  follows from Theorem 3.2.

### 3.2.2 Neumann and Robin boundary conditions

To simplify the explanation, we shall concentrate on the example problem

$$-u'' + c(x)u = f \text{ in } \Omega = (0, 1),$$

with  $c \in L^\infty(0, 1)$  a nonnegative function. We consider boundary conditions of the form

$$-u'(0) + \alpha u(0) = -g_0, \quad u'(1) + \alpha u(1) = g_1$$

which are of Robin-type if  $\alpha > 0$  and Neumann-type if  $\alpha = 0$  and multiplying by a test function  $v \in H^1(0, 1)$  and integration by parts gives

$$\int_0^1 -u'' v + c u v dx = -u'(1)v(1) + u'(0)v(0) + \int_0^1 u' v' + c u v dx = \int_0^1 f v dx$$

We get the weak formulation: Find  $u \in H^1(0, 1)$

$$\int_0^1 u' v' + c u v dx + \alpha u(1)v(1) + \alpha u(0)v(0) = \int_0^1 f v dx + g_1 v(1) - g_0 v(0) \quad (**)$$

for all  $v \in H^1(0, 1)$ .

We shall apply the Lax-Milgram Theorem for the Robin problem here, we see that the equation equation(\*\*) is of the form

$$a(u, v) = \ell(v) \quad \forall v \in V$$

where  $V = H^1(0, 1)$ ,  $\ell(v) = \int_0^1 f v dx + g_1 v(1) - g_0 v(0)$ , and

$$a(u, v) = \int_0^1 u' v' + c u v dx + \alpha u(1)v(1) + \alpha u(0)v(0) dx$$

We now check the assumptions of the Lax-Milgram theorem.

1. (Boundedness of  $\ell(\cdot)$ ). Here since by Theorem 2.20 we have

$$\max_{x \in [0, 1]} |v(x)| \leq C_2 \|v\|_{H^1(0, 1)} \quad \forall v \in H^1(0, 1)$$

for some constant  $C_2 > 0$ . Therefore we have

$$\begin{aligned} |\ell(v)| &\leq \|f\|_{L^2(0,1)} \|v\|_{L^2(0,1)} + (|g_0| + |g_1|)C_2 \|v\|_{H^1(0,1)} \\ &\leq (\|f\|_{L^2(0,1)} + (|g_0| + |g_1|)C_2) \|v\|_{H^1(0,1)} \end{aligned}$$

which shows the boundedness of  $\ell(\cdot)$  on  $H^1(0,1)$ .

2. (Boundedness of  $a(\cdot, \cdot)$ ). Similarly, by Theorem 2.20 and Cauchy-Schwarz inequality we have

$$\begin{aligned} |a(w, v)| &\leq \|w'\|_{L^2(0,1)} \|v'\|_{L^2(0,1)} + \|c\|_{L^\infty(0,1)} \|w\|_{L^2(0,1)} \|v\|_{L^2(0,1)} + \alpha |w(1)| |v(1)| + \alpha |w(0)| |v(0)| \\ &\leq \max_{x \in (0,1)} (1, \|c\|_{L^\infty(0,1)}) \|w\|_{H^1(0,1)} \|v\|_{H^1(0,1)} + 2\alpha C_2^2 \|w\|_{H^1(0,1)} \|v\|_{H^1(0,1)} \end{aligned}$$

which shows boundedness of  $a(\cdot, \cdot)$ .

3. (Coercivity of  $a(\cdot, \cdot)$ ). For  $\alpha > 0$ , we use the nonnegativity of  $c$  to get

$$a(v, v) \geq \|v'\|_{L^2(0,1)}^2 + \alpha |v(1)|^2 + \alpha |v(0)|^2$$

there is a constant  $C_R > 0$  such that

$$\|v\|_{L^2(0,1)}^2 \leq C_R (\|v'\|_{L^2(0,1)}^2 + |v(0)|^2) \quad \forall v \in H^1(0,1)$$

Therefore, after a few further steps, we obtain

$$a(v, v) \geq c_0 \|v\|_{H^1(0,1)}^2$$

with constant  $c_0 = \min(1, \alpha)/(1 + C_R) > 0$ .

**Remark.** It is important to note that terms such as  $u'(0)$  and  $u'(1)$  cannot be left inside the bilinear form, because these terms are not bounded: there is no constant  $C$  such that  $|w'(0)| \leq C \|w\|_{H^1(0,1)}$  for all  $w \in H^1(0,1)$ . This is why we sought to eliminate these terms implicitly using the boundary conditions in the weak formulation of the problem.

**Theorem 3.3.** Let  $c \in L^\infty(0,1)$  be nonnegative and let  $\alpha > 0$ . Then the weak formulation (\*\*) of the boundary problem has a unique weak solution  $u \in H^1(0,1)$ .

### 3.3 Dirichlet's principle and energy minimization

Consider Poisson equation

$$\begin{aligned} \Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned}$$

We want to find  $u \in H_0^1(\Omega)$  such that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega) \quad (*)$$

**Theorem 3.4.** (Dirichlet's principle)  $u \in H_0^1(\Omega)$  is the weak solution of (\*) if and only if  $u$  minimizes the energy functional  $\mathcal{E} : H_0^1(\Omega) \rightarrow \mathbb{R}$  defined by

$$\mathcal{E}(v) = \frac{1}{2} \int_{\Omega} |\nabla v|^2 \, dx - \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega)$$

**Proof.** ( $\implies$ ) Suppose  $u \in H_0^1(\Omega)$  is the weak solution, we need to show that  $\mathcal{E}(u) \leq \mathcal{E}(w)$ ,  $\forall w \in H_0^1(\Omega)$ . Start by  $w = u + v$  for  $v = w - u$ . Then

$$\begin{aligned} \mathcal{E}(w) &= \mathcal{E}(u + v) = \frac{1}{2} \int_{\Omega} |\nabla(u + v)|^2 \, dx - \int_{\Omega} f(u + v) \, dx \\ &= \frac{1}{2} \int_{\Omega} |\nabla u|^2 \, dx - \int_{\Omega} f u \, dx + \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx + \frac{1}{2} \int_{\Omega} |\nabla v|^2 \, dx \\ &= \mathcal{E}(u) + \left[ \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx \right] + \frac{1}{2} \int_{\Omega} |\nabla v|^2 \, dx \\ &\implies \mathcal{E}(u + v) \geq \mathcal{E}(u) \end{aligned}$$

( $\impliedby$ ) Suppose  $u \in H_0^1(\Omega)$  minimizes  $\mathcal{E}(\cdot)$ . Let  $v \in H_0^1(\Omega)$  be arbitrary but fixed. Let  $\delta > 0$ , consider  $v_\delta = u + \delta v$  then

$$\begin{aligned} \mathcal{E}(v_\delta) &= \mathcal{E}(u) + \delta \left[ \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx \right] + \frac{\delta^2}{2} \int_{\Omega} |\nabla v|^2 \, dx \\ \implies 0 &\leq \delta \left[ \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx \right] + \frac{\delta^2}{2} \int_{\Omega} |\nabla v|^2 \, dx \\ \implies 0 &\leq \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx + \frac{\delta}{2} \int_{\Omega} |\nabla v|^2 \, dx \end{aligned}$$

Take limit  $\delta \rightarrow 0$ , we get

$$0 \leq \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx$$

If we repeat the whole thing for  $-v$  then

$$\begin{aligned} 0 &\geq \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx \\ \implies \int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Omega} f v \, dx &= 0 \end{aligned}$$

so  $u$  is a weak solution of  $(*)$ . □

### 3.4 Higher regularity of weak solutions

Often, we want to show that the weak solution of the problem has higher regularity, and in some cases also solves the problem in strong form.

**Example.** Consider

$$-u'' = f \text{ on } (0, 1), \quad f \in L^2(0, 1) \quad (1)$$

where  $u(0) = u(1) = 0$ .

The weak formulation find  $u \in H_0^1(0, 1)$

$$\int_0^1 u' v' \, dx = \int_0^1 f v \, dx \quad \forall v \in H_0^1(0, 1)$$

Claim: If  $f \in L^2(0, 1)$ , then  $u \in H^2(0, 1)$  as well, and its second weak derivative  $u'' = -f$ .

**Proof.** To show  $u \in H^2(0, 1)$ , we must show that  $\exists w \in L^2(0, 1)$  that satisfies the second weak derivative.

$$\int_0^1 u \phi'' \, dx = \int_0^1 w \phi \, dx \quad \forall \phi \in C_0^\infty(0, 1)$$

Let  $\phi \in C_0^\infty(0, 1)$ , note  $\phi' \in C_0^\infty(0, 1)$  then

$$\int_0^1 u \phi'' \, dx = \int_0^1 u (\phi')' \, dx$$

Since  $u \in H^1(0, 1)$ ,

$$\int_0^1 u (\phi')' \, dx = \int_0^1 u' \phi' \, dx = - \int_0^1 f \phi \, dx$$

which shows that  $-f$  satisfies the definition of the second weak derivative of  $u$ , since  $f \in L^2(0, 1)$  we see that  $w = -f \in L^2(0, 1)$  the second weak derivative of  $u \in H^1(0, 1)$  and by definition,  $u'' = -f$ . □

In higher dimensions, if we consider

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned}$$

then the weak solution  $u \in H_0^1(\Omega)$  exists and is unique of all bounded open  $\Omega$ . However, there are cases of domains  $\Omega$  where  $u \notin H^2(0, 1)$ . If  $\Omega \subset \mathbb{R}^n$  is convex set then  $H^2$  regularity of the solution. If  $\Omega \subset \mathbb{R}^n$  domain with smooth boundaries then  $H^2$  regularity of the solution.

## 4 Galerkin's and finite element method

Let  $V$  be a Hilbert space, and let  $a(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$  be bounded and coercive bilinear form. The Lax-Milgram theorem shows that, for any  $\ell \in V^*$ , there exists a unique  $u \in V$  such that

$$a(u, v) = \ell(v) \quad \forall v \in V$$

### 4.1 Galerkin's method

**Definition 4.1.** For integer  $N > 0$ , let  $V_N \subset V$  be a finite dimensional subspace of  $V$  with dimension  $N$ . The Galerkin approximation  $u_N \in V_N$  of  $u \in V$  is defined as the unique solution of

$$a(u_N, v_N) = \ell(v_N) \quad \forall v_N \in V_N.$$

Note that  $u_N \in V_N$  is well defined as result of the Lax-Milgram theorem applied to the restriction of the bilinear form  $a(\cdot, \cdot)$  and linear functional  $\ell(\cdot)$  to  $V_N$ .

(The linear system) The finite dimensionality of  $V_N$  implies that  $u_N$  can be computed in solving a linear system. To show this, let  $\{\phi_i\}_{i=1}^N \subset V_N$  denote a basis of  $V_N$ , i.e.  $\{\phi_i\}_{i=1}^N$  is a linearly independent set for which

$$V_N = \text{span}\{\phi_i\}_{i=1}^N$$

**Theorem 4.2.** Let  $\{\phi_i\}_{i=1}^N$  be a basis of  $V_N$ . Let  $\mathbf{u} = (U_1, \dots, U_N) \in \mathbb{R}^N$  denote the vector of coefficients in the basis expansion of  $u_N$ , i.e.  $u_N = \sum_{j=1}^N U_j \phi_j$ . Let  $\mathbf{A} \in \mathbb{R}^{N \times N}$  and  $\mathbf{f} \in \mathbb{R}^N$  be defined by

$$\mathbf{A}_{ij} = a(\phi_j, \phi_i), \quad \mathbf{f}_i = \ell(\phi_i).$$

Then,  $u_N \in V_N$  solves

$$a(u_N, v_N) = \ell(v_N) \quad \forall v_N \in V_N \quad (*)$$

if and only if  $\mathbf{u} \in \mathbb{R}^N$  solves the linear system

$$\mathbf{A}\mathbf{u} = \mathbf{f}.$$

where  $\mathbf{A}$  is called stiffness matrix and  $\mathbf{f}$  is called load vector.

**Proof.** We start by claiming that  $u_N$  solves (\*) if and only if

$$a(u_N, \phi_i) = \ell(\phi_i) \quad \forall i = 1, \dots, N \quad (**)$$

Starting by supposing that  $u_N$  solves (\*). Then, since  $\phi_i \in V_N$  for each  $i = 1, \dots, N$ , immediately obtain (\*\*). To show the converse, assume that  $u_N$  satisfies (\*\*). Then using the linearity of  $a(\cdot, \cdot)$  and  $\ell(\cdot)$  and the fact that for any  $v_N \in V_N$ , there exists coefficients  $(c_i)_{i=1}^N$  such that  $v_N = \sum_{i=1}^N c_i \phi_i$ . Therefore

$$a(u_N, v_N) = a(u_N, \sum_{i=1}^N c_i \phi_i) = \sum_{i=1}^N c_i a(u_N, \phi_i) = \sum_{i=1}^N c_i \ell(\phi_i) = \ell(\sum_{i=1}^N c_i \phi_i) = \ell(v_N)$$

This shows that (\*) and (\*\*) are equivalent. We now write  $u_N = \sum_{j=1}^N U_j \phi_j$  and use linearity of  $a(\cdot, \cdot)$  again to get  $a(u_N, \phi_i) = \sum_{j=1}^N a(\phi_j, \phi_i) U_j$ . After substitution, we find that (\*) and (\*\*) are both equivalent to

$$\sum_{j=1}^N \mathbf{A}_{ij} U_j = f_i \quad \forall i = 1, \dots, N \Leftrightarrow \mathbf{A}\mathbf{u} = \mathbf{f}.$$

□

#### 4.1.1 Galerkin orthogonality and Céa's Lemma

The Galerkin method is consistent in the sense that

$$a(u, v_N) = \ell(v_N) \quad \forall v_N \in V_N$$

when comparing with (\*). We have

$$a(u - u_N, v_N) = 0 \quad \forall v_N \in V_N$$

which is known as the Galerkin orthogonality. This is not the same as orthogonality in the inner-product  $(\cdot, \cdot)$ . In general,  $a(\cdot, \cdot)$  is not symmetric, so is not an inner-product.

**Lemma 4.3.** (Céa's Lemma) Let  $V$  be a Hilbert space with norm  $\|\cdot\|$ . Let the bilinear form  $a(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$  be bounded

$$|a(w, v)| \leq c_1 \|w\| \|v\| \quad \forall w, v \in V$$

and coercive

$$a(v, v) \geq c_0 \|v\|^2 \quad \forall v \in V$$

Let  $\ell \in V^*$  and  $u$  solve  $a(u, v) = \ell(v)$ ,  $\forall v \in V$ . Let  $V_N \subset V$  be a finite dimensional subspace and  $u_N \in V_N$  solve

$$a(u_N, v_N) = \ell(v_N) \quad \forall v_N \in V_N$$

then

$$\|u - u_N\| \leq \frac{c_1}{c_0} \inf_{v_N \in V_N} \|u - v_N\|$$

**Proof.**

$$\begin{aligned} \|u - u_N\|^2 &\leq \frac{1}{c_0} a(u - u_N, u - u_N) \\ &= \frac{1}{c_0} a(u - u_N, u) \\ &= \frac{1}{c_0} a(u - u_N, u - v_N) \quad \forall v_N \in V_N \end{aligned}$$

so, for any  $v_N \in V_N$ ,

$$\begin{aligned} \|u - u_N\|^2 &\leq \frac{c_1}{c_0} \|u - u_N\| \|u - v_N\| \\ \implies \|u - u_N\| &\leq \frac{c_1}{c_0} \|u - v_N\| \quad \forall v_N \in V_N \end{aligned}$$

since  $v_N$  is arbitrary, proof is completed by taking the infimum.  $\square$

**Corollary 4.4.** Suppose that  $\{V_{N_k}\}_{k=1}^\infty$  is a sequence of finite dimensional subspaces of  $V$  that is dense in the limit:  $\forall v \in V, \exists$  a sequence  $\{v_k\}_{k=1}^\infty$  such that  $v_k \in V_{N_k} \forall k \geq 1, \|v - v_k\| \rightarrow 0$  as  $k \rightarrow \infty$ . Then the Galerkin approximations  $\{u_{N_k}\}_{k=1}^\infty$  converges to the exact solution  $u \in V$ .

**Proof.** Let  $\{V_{N_k}\}_{k=1}^\infty$  satisfy  $\|u - v_k\| \rightarrow 0$  as  $k \rightarrow \infty$ , and  $v_k \in V_{N_k} \forall k \geq 1$ . Then by Céa's Lemma,

$$\|u - u_{N_k}\| \leq \frac{c_1}{c_0} \inf_{v_{N_k} \in V_{N_k}} \|u - v_{N_k}\| \leq \frac{c_1}{c_0} \|u - v_k\|$$

Therefore,

$$\lim_{k \rightarrow \infty} \|u - u_{N_k}\| = 0$$

so  $u_{N_k} \rightarrow u$  as  $k \rightarrow \infty$ .  $\square$

## 4.2 Finite element method

We have seen Galerkin's method for the general variational problems  $a(u, v) = \ell(v), \forall v \in V$ . It is based on choosing  $V_N \subset V$  a finite dimensional subspace.

To apply Galerkin's method to PDE, we will often choose  $V_N$  to be a space of piecewise polynomials in the Sobolev space of the weak form of the PDE: this leads to the idea of finite element method.

(Problems in 1D) Consider

$$\begin{aligned} -u'' + b(x)u' + c(x)u &= f(x) \quad \forall x \in (0, L) \\ u(0) &= 0 = u(L) \end{aligned}$$

The weak formulation, find  $u \in H_0^1(0, L)$  such that

$$\int_0^L u'v' + b(x)u'v + c(x)uv \, dx = \int_0^L f(x)v \, dx \quad \forall v \in H_0^1(0, L)$$

We have seen before that if  $b \in W^{1,\infty}(0, L)$ ,  $c \in L^\infty(0, L)$ ,  $c - \frac{1}{2}b' \geq 0$ . Then  $a(\cdot, \cdot)$  is bounded and coercive on  $H_0^1(0, L)$ .

To build a finite dimensional subspace, let us partition  $[0, L]$  into sub-intervals  $\{[x_{i-1}, x_i]\}_{i=1}^{N+1}$  for some  $N \geq 1$ . Assume that all intervals have equal length in particular  $x_i - x_{i-1} = h = \frac{L}{N+1}$ . The nodes  $x_0, x_{N+1}$  are called boundary nodes.

The  $\{x_i\}_{i=1}^N$  are called interior nodes. For each interior node  $\{x_i\}_{i=1}^N$ , define the basis(hat) function by

$$\phi_i(x) = \max\left(1 - \frac{|x - x_i|}{h}, 0\right) = \begin{cases} 1 + \frac{x - x_i}{h} & x \in [x_{i-1}, x_i] \\ 1 - \frac{x - x_i}{h} & x \in [x_i, x_{i+1}] \\ 0 & \text{otherwise} \end{cases}$$

The functions  $\{\phi_i\}_{i=1}^N$  satisfy:

- (1) Continuous on  $[0, L]$ , piecewise affine with respect to  $\{[x_{i-1}, x_i]\}_{i=1}^N$ .
- (2) Nodal.  $\phi_i(x_j) = \delta_{ij} \forall j = 0, \dots, N+1$ .
- (3) Conforming.  $\phi_i \in H_0^1(0, L)$  for each  $i = 1, \dots, N$ .
- (4) Support. Support of  $\phi_i$  only overlaps with  $\phi_{i-1}$  and  $\phi_{i+1}$ .

We now define the finite element space

$$V_h = \text{Span}\{\phi_i\}_{i=1}^N$$

then  $\dim V_h = N \geq 1$ .

**Lemma 4.5.** The finite element space  $V_h$  consists of all continuous piecewise affines (polynomials of degree at most one on each elements) with respect to  $\{[x_{i-1}, x_i]\}_{i=1}^N$  that are in  $H_0^1(0, L)$ .

In particular, since  $\{\phi_i\}_{i=1}^N$  are nodal basis  $v_h \in V_h$  then  $v_h = \sum_{i=1}^N W_i \phi_i \Leftrightarrow W_i = v_h(x_i) \forall i = 1, \dots, N$ .

**Definition 4.6.** The finite element approximation of  $a(u, v) = \int_0^L f v \, dx \forall v \in H_0^1(0, L)$  is then defined by Galerkin's method using  $V_h$ : find  $u_h \in V_h$  such that

$$a(u_h, v_h) = \int_0^L f v_h \, dx \quad \forall v_h \in V_h \quad (*)$$

Since  $a(\cdot, \cdot)$  is bounded and coercive on  $H_0^1(0, L)$  and since  $v_h \subset H_0^1(0, L)$  is closed in  $H_0^1(0, L)$ ,  $a(\cdot, \cdot)$  also satisfies the conditions of Lax-Milgram theorem, so there exists a unique  $u_h \in V_h$  solving  $(*)$ . In addition,

$$a(u_h, v_h) = \int_0^L u_h' v_h' + b u_h' v_h + c u_h v_h \, dx$$

is well-defined for  $u_h, v_h \in V_h$  by weak formulation leads to terms only with first weak derivatives. NB:  $v_h \in V_h$  are not twice weakly differentiable.

**Example.** Consider

$$\begin{aligned} -u'' + u' &= f(x) \\ u(0) &= u(L) = 0 \end{aligned}$$

We saw previously that  $a(u_h, v_h) = \ell(v_h)$ ,  $\forall v_h \in V_h$  if and only if  $A\mathbf{u} = \mathbf{f}$  with  $A_{ij} = a(\phi_j, \phi_i)$ . The bilinear form is

$$a(u_h, v_h) = \int_0^L u_h' v_h' + u_h' v_h \, dx$$

and

$$A_{ij} = \int_0^L \phi_j' \phi_i' + \phi_j' \phi_i \, dx \quad i, j = 1, \dots, N$$

$\Rightarrow A_{ij=0}$  if  $j \notin \{i-1, i, i+1\}$ . Consider  $j = i$  then

$$\begin{aligned} A_{ii} &= \int_0^L (\phi_i')^2 + \phi_i' \phi_i \, dx \\ &= \int_{x_{i-1}}^{x_i} (\phi_i')^2 + \phi_i' \phi_i \, dx + \int_{x_i}^{x_{i+1}} (\phi_i')^2 + \phi_i' \phi_i \, dx \\ &= \int_{x_{i-1}}^{x_i} \left(\frac{1}{h}\right)^2 + \frac{1}{h} \left(1 + \frac{x - x_i}{h}\right) \, dx + \int_{x_i}^{x_{i+1}} \left(-\frac{1}{h}\right)^2 - \frac{1}{h} \left(1 - \frac{x - x_i}{h}\right) \, dx \\ &= \frac{1}{h} + \frac{1}{h} \frac{h}{2} + \frac{1}{h} - \frac{1}{2} = \frac{2}{h} \end{aligned}$$

Consider  $j = i-1$  then

$$\begin{aligned} A_{i,i-1} &= \int_0^L \phi_{i-1}' \phi_i + \phi_{i-1}' \phi_i \, dx \\ &= \int_{x_{i-1}}^{x_i} \left(-\frac{1}{h}\right) \frac{1}{h} + \left(-\frac{1}{h}\right) \left(1 + \frac{x - x_i}{h}\right) \, dx \\ &= -\frac{1}{h} - \frac{1}{2} \end{aligned}$$

And similarly,  $A_{i,i+1} = -\frac{1}{h} + \frac{1}{2}$  thus,

$$A = \frac{1}{h} \begin{pmatrix} 2 & -1 & \dots & \dots \\ -1 & 2 & -1 & \dots \\ 0 & -1 & 2 & \dots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} + \begin{pmatrix} 0 & 1/2 & \dots & \dots \\ -1/2 & 0 & 1/2 & \dots \\ 0 & -1/2 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

The linear system  $A\mathbf{u} = \mathbf{f}$  then

$$\begin{aligned} \frac{-U_{i-1} + 2U_i - U_{i+1}}{h} + \frac{U_{i+1} - U_{i-1}}{2} &= \int_0^L f \phi_i \, dx \\ \Rightarrow \frac{-U_{i-1} + 2U_i - U_{i+1}}{h^2} + \frac{U_{i+1} - U_{i-1}}{2h} &= \frac{1}{h} \int_0^L f \phi_i \, dx \approx f(x_i) \end{aligned}$$

## 4.3 A priori error bounds

Recall Céa's lemma

$$\|u - u_h\|_{H^1(0,L)} \leq \frac{c_1}{c_0} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(0,L)}$$

Our goal now is to quantify  $\inf_{v_h \in V_h} \|u - v_h\|_{H^1(0,L)}$  as a function of the size of the mesh elements  $h = |x_i - x_{i-1}|$ .

**Lemma 4.7.** Suppose that  $u \in H_0^1(0, L) \cap H^2(0, L)$ . Let  $I_h u \in V_h$  be defined as the interpolant of  $u$  on  $V_h$  at the nodes  $I_h u \in V_h$  and  $I_h u(x_i) = u(x_i)$ ,  $\forall i = 0, \dots, N+1$ .

$$\begin{aligned}\|u - I_h u\|_{L^2(x_{i-1}, x_i)} &\leq \frac{h^2}{\pi^2} \|u''\|_{L^2(x_{i-1}, x_i)} \\ \|(u - I_h u)'\|_{L^2(x_{i-1}, x_i)} &\leq \frac{h}{\pi} \|u''\|_{L^2(x_{i-1}, x_i)}\end{aligned}$$

**Proof.** Define the function  $z = u - I_h u \implies z(x_i) = 0$  so  $z|_{(x_{i-1}, x_i)} \in H_0^1(x_{i-1}, x_i) \cap H^2(x_{i-1}, x_i)$ . The function  $z$  has the Fourier sine series expansion

$$z|_{(x_{i-1}, x_i)}(x) = \sum_{k=1}^{\infty} a_k \sin\left(\frac{\pi k}{h}(x - x_{i-1})\right) \quad \forall x \in [x_{i-1}, x_i]$$

Then

$$\begin{aligned}\|z\|_{L^2(x_{i-1}, x_i)}^2 &= \frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 \\ z' &= \sum_{k=1}^{\infty} a_k \frac{\pi k}{h} \cos\left(\frac{\pi k}{h}(x - x_{i-1})\right) \\ \implies \|z'\|_{L^2(x_{i-1}, x_i)}^2 &= \frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 \frac{\pi^2 k^2}{h^2}\end{aligned}$$

Similarly,

$$\|z''\|_{L^2(x_{i-1}, x_i)}^2 = \frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 \left(\frac{\pi k}{h}\right)^4$$

For  $k \geq 1$ ,  $k^4 \geq k^2 \geq 1$  so

$$\|z\|_{L^2(x_{i-1}, x_i)}^2 \leq \frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 k^4 = \frac{h^4}{\pi^4} \left(\frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 \left(\frac{\pi k}{h}\right)^4\right) = \frac{h^4}{\pi^4} \|z''\|_{L^2(x_{i-1}, x_i)}^2$$

In addition,

$$\|z'\|_{L^2(x_{i-1}, x_i)}^2 = \frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 \left(\frac{\pi k}{h}\right)^2 \leq \frac{h^2}{\pi^2} \left(\frac{h}{2} \sum_{k=1}^{\infty} |a_k|^2 \left(\frac{\pi k}{h}\right)^4\right) = \frac{h^2}{\pi^2} \|z''\|_{L^2(x_{i-1}, x_i)}^2$$

Recall  $I_h u \in V_h$  so  $I_h u|_{(x_{i-1}, x_i)}$  is a polynomial of degree at most 1 so  $(I_h u|_{(x_{i-1}, x_i)})'' = 0$

$$\begin{aligned}\implies \|z''\|_{L^2(x_{i-1}, x_i)} &= \|u''\|_{L^2(x_{i-1}, x_i)} \\ \implies \|u - I_h u\|_{L^2(x_{i-1}, x_i)} &\leq \frac{h^2}{\pi^2} \|u''\|_{L^2(x_{i-1}, x_i)} \\ \|(u - I_h u)'\|_{L^2(x_{i-1}, x_i)} &\leq \frac{h}{\pi} \|u''\|_{L^2(x_{i-1}, x_i)}\end{aligned}$$

□

**Theorem 4.8.** Let  $u_h \in V_h$  be the Galerkin approximation of  $u$ , suppose that  $u \in H^2(0, L)$ . Then

$$\|u - u_h\|_{H^1(0, L)} \leq \frac{c_1}{c_0} \frac{h}{\pi} \sqrt{1 + \frac{h^2}{\pi^2}} \|u''\|_{L^2(0, L)}$$

**Proof.** Clearly,

$$\begin{aligned}\inf_{v_h \in V_h} \|u - v_h\|_{H^1(0, L)} &\leq \|u - I_h u\|_{H^1(0, L)} = (\|u - I_h u\|_{L^2(0, L)}^2 + \|(u - I_h u)'\|_{L^2(0, L)}^2)^{1/2} \\ &= \left(\sum_{i=1}^{N+1} \|u - I_h u\|_{L^2(x_{i-1}, x_i)}^2 + \|(u - I_h u)'\|_{L^2(x_{i-1}, x_i)}^2\right)^{1/2} \\ &\leq \left(\sum_{i=1}^{N+1} \left(\frac{h^4}{\pi^4} \|u''\|_{L^2(x_{i-1}, x_i)}^2 + \frac{h^2}{\pi^2} \|u''\|_{L^2(x_{i-1}, x_i)}^2\right)\right)^{1/2} \\ &= \left(\frac{h^2}{\pi^2} \left(1 + \frac{h^2}{\pi^2}\right) \sum_{i=1}^{N+1} \|u''\|_{L^2(x_{i-1}, x_i)}^2\right)^{1/2} \\ &= \left(\frac{h^2}{\pi^2} \left(1 + \frac{h^2}{\pi^2}\right) \|u''\|_{L^2(0, L)}^2\right)^{1/2} \\ \implies \|u - I_h u\|_{H^1(0, L)} &\leq \frac{h}{\pi} \sqrt{1 + \frac{h^2}{\pi^2}} \|u''\|_{L^2(0, L)} \\ \implies \|u - u_h\|_{H^1(0, L)} &\leq \frac{c_1}{c_0} \frac{h}{\pi} \sqrt{1 + \frac{h^2}{\pi^2}} \|u''\|_{L^2(0, L)}\end{aligned}$$

□

**Exercise.** (Q2 Exam 2023) Consider

$$-u'' = f \quad x \in (0, 1), f \in L^2(0, 1)$$

subjected to the initial conditions  $u(0) = P$ ,  $u(1) = Q$  and  $P, Q \in \mathbb{R} \setminus \{0\}$ . Define mesh  $\mathcal{T}_h = \{[x_{i-1}, x_i]\}_{i=1}^{N+1}$ ,  $h = \frac{1}{N+1}$  mesh of subintervals of equal length  $h$ .  $V_h$  is space of functions in  $H_0^1(0, 1)$  that are piecewise affine with respect to  $\mathcal{T}_h$ . Let  $u_h$  is the function in  $H^1(0, 1)$  that is piecewise affine with respect to  $\mathcal{T}_h$  that solves

$$\int_0^1 u'_h v'_h dx = \int_0^1 f v_h dx, \quad \forall v_h \in V_h, u_h(0) = P, u_h(1) = Q$$

Show that  $\|u - u_h\|_{H^1(0,1)} \leq Ch\|u''\|_{L^2(0,1)}$  for some constants to be determined.

**Proof.** Let  $\tilde{u} = P(1-x) + Qx$  which satisfies  $\tilde{u}(0) = P$  and  $\tilde{u}(1) = Q$ . Let  $u = \tilde{u} + u_0$ ,  $u_0 \in H_0^1(0, 1)$  satisfies

$$\begin{aligned} a(u_0, v) &= \int_0^1 u'_0 v'_0 dx = \int_0^1 f v dx - a(\tilde{u}, v), \quad \forall v \in H_0^1(0, 1) \\ \implies a(u, v) &= a(u_0, v) + a(\tilde{u}, v) = \int_0^1 f v dx, \quad \forall v \in H_0^1(0, 1) \end{aligned}$$

then

$$\begin{aligned} a(u, v) &= \int_0^1 f v dx \quad \forall v \in H_0^1(0, 1) \\ a(u_h, v_h) &= \int_0^1 f v_h dx \quad \forall v_h \in V_h \end{aligned}$$

Note  $u, u_h \notin H_0^1(0, 1)$  so  $u_h \notin V_h$ , however  $(u - u_h)(0) = 0$  and  $(u - u_h)(1) = 0 \implies u - u_h \in H_0^1(0, 1)$ . Since  $\forall v_h \in V_h \subset H_0^1(0, 1)$  so  $a(u - u_h, v_h) = 0$ . Thus, by coercivity

$$\begin{aligned} \|u - u_h\|_{H^1(0,1)}^2 &\leq \frac{1}{c_0} a(u - u_h, u - u_h) \\ &= \frac{1}{c_0} a(u - u_h, u - u_h - v_h) \\ &\leq \frac{c_1}{c_0} \|u - u_h\|_{H^1(0,1)} \|u - u_h - v_h\|_{H^1(0,1)} \\ \implies \|u - u_h\|_{H^1(0,1)} &\leq \frac{c_1}{c_0} \|u - u_h - v_h\|_{H^1(0,1)} \quad \forall v_h \in V_h \\ \implies \|u - u_h\|_{H^1(0,1)} &\leq \frac{c_1}{c_0} \inf_{v_h \in V_h} \|u - u_h - v_h\|_{H^1(0,1)} \end{aligned}$$

Assume now  $u \in H^2(0, 1)$ ,  $v_h = I_h(u - u_h)$  then  $v_h \in V_h$  since  $I_h(u - u_h)(0) = (u - u_h)(0) = 0$ ,  $I_h(u - u_h)(1) = 0$ . Claim

$$u - u_h - I_h(u - u_h) = u - I_h(u)$$

It suffices to show that  $u_h = I_h(u_h)$ .  $u_h$  is  $H^1(0, 1)$  continuous piecewise affine function with respect to  $\mathcal{T}_h$ . Then  $u_h - I_h(u_h)$  is  $H^1(0, 1)$  continuous piecewise affine, and  $(u_h - I_h(u_h))(x_i) = 0$ ,  $\forall i = 0, \dots, N+1$ . In this case,  $u_h - I_h(u_h)$  vanishes at all nodes, also in  $H_0^1(0, 1)$  and thus  $u_h - I_h u_h \in V_h = \text{Span}\{\phi_i\}_{i=1}^N$ .

$$\begin{aligned} \implies u_h - I_h u_h &= \sum_{i=1}^N c_i \phi_i, \quad c_i = u_h(x_i) - I_h u_h(x_i) = 0 \quad i = 1, \dots, N \\ \implies u_h - I_h u_h &= 0 \end{aligned}$$

$$\implies \|u - u_h\|_{H^1(0,1)} \leq \frac{c_1}{c_0} \|u - u_h - I_h(u - u_h)\|_{H^1(0,1)} = \frac{c_1}{c_1} \|u - I_h u\|_{H^1(0,1)} \leq \frac{c_1}{c_0} \frac{h}{\pi} \sqrt{1 + \frac{h^2}{\pi^2}} \|u''\|_{L^2(0,1)}$$

In particular,

$$C = \frac{c_1}{c_0} \frac{1}{\pi} \sqrt{1 + \frac{1}{\pi^2}}$$

further calculations can be done to derive  $c_0 = \frac{2}{3}$  and  $c_1 = 1$  so

$$C = \frac{3}{2\pi} \sqrt{1 + \frac{1}{\pi^2}}$$

□

## 4.4 Other boundary conditions

To handle other boundary conditions (Neumann/Robin), the main idea is to adjust the choice of the finite elements space  $V_h$  to match the solution space  $V$  of the weak form of the problem. General rule is: choose  $V_h$  consists of all piecewise affine functions on  $\mathcal{T}_h$  that belong to the solution space  $V$  of the problem.

**Example.** Consider

$$-u'' + u' = f, \quad x \in (0, 1)$$

subjected to the initial condition  $u(0) = 0$ ,  $u'(1) = 0$  and  $f \in L^2(0, 1)$ .

The weak form is to find  $u \in H_{E_0}^1(0, 1)$  such that

$$\int_0^1 u'v' + u'v \, dx = \int_0^1 f v \, dx, \quad \forall v \in H_{E_0}^1(0, 1)$$

Denote the LHS of the equation by  $a(u, v)$  and the RHS of the equation by  $\ell(v)$ . To show the boundedness, note that

$$|a(w, v)| \leq \|w'\|_{L^2} \|v'\|_{L^2} + \|w'\|_{L^2} \|v\|_{L^2} \leq 2\|w\|_{H^1} \|v\|_{H^1} \quad \forall w, v \in H_{E_0}^1(0, 1)$$

To show the coercivity,

$$a(v, v) = \int_0^1 |v'|^2 + v'v \, dx = \int_0^1 |v'|^2 \, dx + \frac{1}{2}|v(1)|^2 \geq \frac{2}{3}\|v\|_{H^1}^2, \quad \forall v \in H_{E_0}^1(0, 1)$$

More generally,  $\mathcal{T}_h = \{[x_{i-1}, x_n]\}_{i=1}^{N+1}$ ,  $0 = x_0 \leq x_1 \leq \dots \leq x_{N+1} = 1$ , then  $V_h = \text{Span}\{\phi_i\}_{i=1}^{N+1}$ . The FE approximation find  $u_h \in V_h$  such that  $a(u_h, v_h) = \ell(v_h)$ ,  $\forall v_h \in V_h$ . Calculation of stiffness matrix:  $A_{ij} = a(\phi_j, \phi_i)$  where calculations is similar to previous example for Dirichlet condition in the case  $i, j \in \{1, \dots, N\}$  which is different when  $i, j$  are equal to  $N+1$ . If  $i, j \in \{1, \dots, N\}$  then

$$A_{ij} = \begin{cases} -\frac{1}{h} - \frac{1}{2} & \text{if } j = i-1 \\ \frac{2}{h} & \text{if } i = j \\ -\frac{1}{h} + \frac{1}{2} & \text{if } j = i+1 \end{cases}$$

$$\Rightarrow \frac{-U_{i-1} + 2U_i - U_{i+1}}{h} + \frac{U_{i+1} - U_{i-1}}{2} = \int_0^1 f \phi_i \, dx$$

To need to calculate  $A_{(N+1),j}$  and note that  $A_{(N+1),j} = 0 \, \forall j < N$  because supports of  $\phi_j$  and  $\phi_{N+1}$  do not overlap for a set of positive measure. Then

$$A_{(N+1),N} = \int_0^1 \phi'_N \phi'_{N+1} + \phi'_N \phi_{N+1} \, dx$$

$$= \int_{x_N}^{x_{N+1}} \left(-\frac{1}{h}\right)\left(\frac{1}{h}\right) - \frac{1}{h}\left(1 + \frac{x - x_{N+1}}{h}\right) \, dx$$

$$= -\frac{1}{h} - \frac{1}{2}$$

Similarly,

$$A_{(N+1),(N+1)} = \int_0^1 (\phi'_{N+1})^2 + \phi'_{N+1} \phi_{N+1} \, dx$$

$$= \int_{x_N}^{x_{N+1}} \left(\frac{1}{h}\right)^2 + \frac{1}{h}\left(1 + \frac{x - x_{N+1}}{h}\right) \, dx$$

$$= \frac{1}{h} + \frac{1}{2}$$

$$\Rightarrow \frac{U_{N+1} - U_N}{h} + \frac{1}{2}(U_{N+1} - U_N) = \int_0^1 f \phi_{N+1} \, dx$$

**Exercise.** (Q4 2019) Consider

$$-u'' = f, \quad x \in (0, 1)$$

subjected to the initial condition  $u(0) = 0$ ,  $u'(1) + u(1) = g$ ,  $f \in L^2(0, 1)$ .

(a) Find the weak formulation for the equation and determine the Galerkin approximation for this.

(b) Show that there exists some constant  $C_1$  such that

$$\|u - u_h\|_{H^1(0,1)} \leq C_1 \inf_{v_h \in V_h} \|u - v_h\|_{H^1(0,1)}$$

(c) Show if  $u \in H^2(0, 1)$  then there exists some constant  $C_2$  such that

$$\|u - u_h\|_{H^1} \leq C_2 h \|u\|_{H^2(0,1)}$$

(a) We find the weak formulation for  $u \in H_{E_0}^1(0, 1)$  where

$$\int_0^1 -u''v \, dx = [-u'v]_0^1 + \int_0^1 u'v' \, dx = (u(1) - g)v(1) + \int_0^1 u'v' \, dx, \quad \forall v \in v \in H_{E_0}^1(0, 1)$$

So we find  $u \in H_{E_0}^1(0, 1)$  such that

$$a(u, v) = \int_0^1 u'v' \, dx + u(1)v(1) = \int_0^1 f v \, dx + g v(1) = \ell(v), \quad \forall v \in H_{E_0}^1(0, 1)$$

FE approximation: find  $u_h \in V_h$  such that  $a(u_h, v_h) = \ell(v_h)$ ,  $\forall v_h \in V_h$ .  $V_h \subset H_{E_0}^1(0, 1) \implies a(u, v_h) = \ell(v_h)$ ,  $\forall v_h \in V_h$  which means  $a(u, v_h) - a(u_h, v_h) = 0$ ,  $\forall v_h \in V_h$ .

(b) We look at the boundedness and coercivity of  $a(\cdot, \cdot)$ . For the boundedness,

$$|a(w, v)| = \left| \int_0^1 w'v' \, dx + w(1)v(1) \right| \leq \left| \int_0^1 w'v' \, dx \right| + |w(1)||v(1)|$$

$$\leq \|w'\|_{L^2} \|v'\|_{L^2} + \|w'\|_{L^2} \|v'\|_{L^2} = 2\|w'\|_{L^2} \|v'\|_{L^2}$$

$$\leq 2\|w\|_{H^1} \|v\|_{H^1} \quad \forall w, v \in H_{E_0}^1(0, 1)$$

For the coercivity,

$$\begin{aligned} a(v, v) &= \int_0^1 |v'|^2 dx + |v(1)| \leq \int_0^1 |v'|^2 dx \\ &\leq \frac{2}{3} \|v\|_{H^1}^2 \quad \forall v \in H_{E_0}^1(0, 1) \end{aligned}$$

Then

$$\begin{aligned} \|u - u_h\|_{H^1}^2 &\leq \frac{3}{2} a(u - u_h, u - u_h) = \frac{3}{2} (a(u - u_h, u) - a(u - u_h, u_h)) \\ &= \frac{3}{2} a(u - u_h, u) = \frac{3}{2} a(u - u_h, u - v_h) \quad \forall v_h \in V_h \\ \implies \|u - u_h\|_{H^1}^2 &\leq \frac{3}{2} \cdot 2 \|u - u_h\|_{H^1} \|u - v_h\|_{H^1} \\ \implies \|u - u_h\| &\leq 3 \|u - v_h\|_{H^1} \quad \forall v_h \in V_h \end{aligned}$$

Take the infimum yields

$$\|u - u_h\|_{H^1} \leq 3 \inf_{v_h \in V_h} \|u - v_h\|_{H^1}$$

i.e.  $C_1 = 3$  satisfies the inequality.

(c) We use

$$\|u - I_h u\|_{H^1} \leq \frac{h}{\pi} \sqrt{1 + \frac{h^2}{\pi^2}} \|u''\|_{L^2(0,1)}$$

Recall  $V_h = \text{Span}\{\phi_i\}_{i=1}^{N+1}$  where  $I_h u = \sum_{i=1}^{N+1} u(x_i) \phi_i \in V_h$ . Since  $u \in H_{E_0}^1(0, 1)$  we have  $u(x_0) = 0 = I_h u(x_0) \implies u(x_i) = I_h u(x_i), \forall i = 0, \dots, N+1$ . Therefore, from previous results

$$\begin{aligned} \|u - I_h u\|_{L^2(x_{i-1}, x_i)} &\leq \frac{h^2}{\pi^2} \|u''\|_{L^2(x_{i-1}, x_i)} \\ \|(u - I_h u)'\|_{L^2(x_{i-1}, x_i)} &\leq \frac{h}{\pi} \|u''\|_{L^2(x_{i-1}, x_i)} \end{aligned}$$

By part(b) we have

$$\|u - u_h\|_{H^1} \leq 3 \|u - I_h u\|_{H^1} \leq 3 \frac{h}{\pi} \sqrt{1 + \frac{h^2}{\pi^2}} \|u''\|_{L^2(0,1)} \leq \frac{3}{\pi} \sqrt{1 + \frac{1}{\pi^2}} h \|u\|_{H^2(0,1)}$$

where  $C_2 = \frac{3}{\pi} \sqrt{1 + \frac{1}{\pi^2}}$ .

# 5 Further topics

## 5.1 Tow dimensional problems

$$-\sum_{i,j=1}^2 \frac{\partial}{\partial x_i} (a_{ij} \frac{\partial u}{\partial x_j}) + \sum_{i,j=1}^2 b_i \frac{\partial u}{\partial x_i} + cu = f \quad \text{on } \Omega \subset \mathbb{R}^2 \quad (2)$$

and  $u = 0$  on  $\partial\Omega$ . Consider a partition of  $\Omega$  into a finite number of closed triangles  $T$  such that any two triangles either have zero intersection, or intersect only at a vertex or a whole angle. The collection of such triangles is called mesh  $\mathcal{T}_h$  and  $h = \max_{T \in \mathcal{T}_h} \text{diam } T$  known as the mesh-size. Let  $\{x_i\}_{i=1}^N$  denote the set of all interior vertices  $\mathcal{T}_h$  i.e.  $x_i$  are vertices of triangles in  $\mathcal{T}_h$  and  $x_i \in \Omega$ . (ordering of  $\{x_i\}$  is not important). For each  $x_i$ ,  $i = 1, \dots, N$ . Define  $\phi_i$  as the unique function satisfies

1.  $\phi_i$  is continuous on  $\overline{\Omega}$ .
2.  $\phi_i$  is piecewise affine with respect to  $\mathcal{T}_h$  i.e.

$$\phi_i|_T(x, y) = a_T x + b_T y + c_T, \quad \forall (x, y) \in T$$

where  $a_T, b_T, c_T$  are constants.

3.  $\phi_i$  is equal to 1 at  $x_i$ , and 0 at all other vertices.

For equation(1), we define FE space  $V_h = \text{span}\{\phi_i\}_{i=1}^N$ , and the FE approximation of the solution of (1) is defined by finding  $u_h \in V_h$  such that

$$\int_{\Omega} a_{ij} \sum_{i,j} \frac{\partial u_h}{\partial x_j} \frac{\partial v_h}{\partial x_i} + \sum_i b_i \frac{\partial u_h}{\partial x_i} v_h + cu_h v_h \, d\mathbf{x} = \int_{\Omega} f v_h \, d\mathbf{x}, \quad \forall v_h \in V_h \quad (3)$$

Recall from chapter 3 for sufficient conditions on  $a_{ij}, b_i, c, f$  to ensure that the bilinear form and RHS satisfy the conditions of Lax-Milgram theorem. Then, since  $\phi_i \in H_0^1(\Omega)$ ,  $\forall i = 1, \dots, N$  so  $V_h \subset H_0^1(\Omega)$  is closed in  $H_0^1(\Omega)$ . So LMT also generated existence of uniqueness of  $u_h$  solution of (2) whenever bilinear form bounded and coercive on  $H_0^1(\Omega)$  and RHS bounded on  $H_0^1(\Omega)$ .

**Example.** Let  $\Omega = (0, 1)^2$ , consider  $\Omega$  to be partitioned into a mesh  $\mathcal{T}_h$  that is formed by first taking a square grid of side length  $h_e = \frac{1}{k+1}$  where  $\mathbb{Z} \ni k \geq 1$ , and then subdividing each square into triangle along its top-left diagonal. Let  $x_{ij}$  denote the interior vertices,  $x_{ij} = (ih_e, jh_e)$ ,  $i, j = 1, \dots, k$ . Total of  $N = k^2$ . Then each basis function  $\phi_{ij}$  corresponding to  $x_{ij}$  can be expressed as follows then

$$\phi_{ij} = \begin{cases} 1 - \frac{x-x_i}{h_e} - \frac{y-y_j}{h_e} & \text{on } T_1 \\ 1 - \frac{y-y_j}{h_e} & \text{on } T_2 \\ 1 + \frac{x-x_i}{h_e} & \text{on } T_3 \\ 1 + \frac{x-x_i}{h_e} + \frac{y-y_j}{h_e} & \text{on } T_4 \\ 1 + \frac{y-y_j}{h_e} & \text{on } T_5 \\ 1 - \frac{x-x_i}{h_e} & \text{on } T_6 \end{cases}$$

**Example.** Consider the PDE

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned}$$

then FE approximation is: find  $u_h \in V_h$  such that

$$\begin{aligned} \int_{\Omega} \nabla u_h \cdot \nabla v_h \, d\mathbf{x} &= \int_{\Omega} f v_h \, d\mathbf{x}, \quad \forall v_h \in V_h \\ \Leftrightarrow A \underline{u} &= f, \quad (A_{(ij)(kl)}) \\ A_{(ij)(kl)} &= \int_{\Omega} \nabla \phi_{ij} \cdot \nabla \phi_{kl} \, d\mathbf{x} \end{aligned}$$

In general, at most 7 choice of  $kl$  such that  $A_{(ij)(kl)}$  is potentially nonzero. We calculate  $ij = kl$  then

$$A_{(ij)(ij)} = \int_{\Omega} |\nabla \phi_{ij}|^2 \, d\mathbf{x} = \sum_{M=1}^6 \int_{\mathcal{T}_M} |\nabla \phi_{ij}|^2 \, d\mathbf{x}$$

note that  $\nabla \phi_{ij}$  is

$$\nabla \phi_{ij} = \frac{1}{h_e} \begin{cases} (-1, -1) & \text{on } T_1 \\ (0, -1) & \text{on } T_2 \\ (1, 0) & \text{on } T_3 \\ (1, 1) & \text{on } T_4 \\ (0, 1) & \text{on } T_5 \\ (1, 0) & \text{on } T_6 \end{cases}$$

then

$$\sum_{M=1}^6 \int_{\mathcal{T}_M} |\nabla \phi_{ij}|^2 dx = \frac{1}{2}(2 + 1 + 1 + 2 + 1 + 1) = 4$$

Then we look at  $A_{(ij),(i+1,j)}$

$$\begin{aligned} A_{(ij),(i+1,j)} &= \int_{\Omega} \nabla \phi_{ij} \cdot \nabla \phi_{(i+1,j)} dx \\ &= \int_{T_1} \nabla \phi_{ij} \cdot \nabla \phi_{(i+1,j)} dx + \int_{T_6} \nabla \phi_{ij} \cdot \nabla \phi_{(i+1,j)} dx \\ &= \frac{h_e^2}{2} \frac{1}{h_e} \begin{pmatrix} -1 \\ -1 \end{pmatrix} \cdot \frac{1}{h_e} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{h_e^2}{2} \frac{1}{h_e} \begin{pmatrix} -1 \\ 0 \end{pmatrix} \cdot \frac{1}{h_e} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= -\frac{1}{2} - \frac{1}{2} = -1 \end{aligned}$$

where we have used the following results

$$\nabla \phi_{(i+1,j)}|_{T_1} = \nabla \phi_{ij}|_{T_3} = \frac{1}{h_e} (1, 0)$$

$$\nabla \phi_{(i+1,j)}|_{T_6} = \nabla \phi_{ij}|_{T_4} = \frac{1}{h_e} (1, 1)$$

Similarly,

$$A_{(ij)((i-1)j)} = A_{(ij)((i+1)j)} = -1$$

Linear system  $a(u_h, v_h) = \int_{\Omega} f v_h dx$  is equivalent to

$$-U_{(i-1)j} - U_{(i+1)j} + 4U_{ij} - U_{i(j-1)} - U_{i(j+1)} = \int_{\Omega} f \phi_{ij} dx, \forall i = 1, \dots, k$$

## 5.2 Rates of convergence

Consider the following equations

$$-\sum_{i,j=1}^2 \frac{\partial}{\partial x_i} (a_{ij} \frac{\partial u}{\partial x_j}) + \sum_{i=1}^2 b_i \frac{\partial u}{\partial x_i} + cu = f \text{ in } \Omega$$

$$u = 0 \text{ on } \partial\Omega$$

where  $a_{ij} = a_{ji}$ ,  $\sum_{i,j=1}^2 a_{ij} v_i v_j \geq c_* |v|^2$ ,  $\forall v = (v_1, v_2)$ . And  $b_i \in W^{1,\infty}$ ,  $c \in L^\infty$ ,  $c - \frac{1}{2} \sum_{i=1}^2 \frac{\partial b_i}{\partial x_i} \geq 0$ . Then the weak form of the equation is to find  $u \in H_0^1(\Omega)$  such that

$$a(u, v) = \int_{\Omega} \sum_{i,j=1}^2 a_{ij} \frac{\partial u}{\partial x_j} \frac{\partial v}{\partial x_i} + \sum_{i=1}^2 b_i \frac{\partial u}{\partial x_i} v + cuv dx = \ell(v) = \int_{\Omega} f v dx \quad \forall v \in H_0^1(\Omega)$$

Consider a sequence of meshes  $\{\mathcal{T}_h\}_{h \in \mathcal{H}}$ , where the set  $\mathcal{H}$  is a sequence of mesh sizes that tends to zero i.e.  $\mathcal{H} = \{h_j\}_{j=1}^\infty$ ,  $h_j \rightarrow 0$  as  $j \rightarrow \infty$ . We assume that each  $\mathcal{T}_h$  is conforming, two elements in  $\mathcal{T}_h$  intersect at a vertex, whole angle, or not at all. For each  $h \in \mathcal{H}$ , let  $V_h \subset H_0^1(\Omega)$  be the usual FE space. Since  $a(\cdot, \cdot)$  is coercive, bounded, we have Céa's lemma

$$\|u - u_h\|_{H^1(\Omega)} \leq \frac{c_1}{c_0} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)}$$

**Definition 5.1.** We say that  $\{\mathcal{T}_h\}_{h \in \mathcal{H}}$  is a shape regular if  $\exists C > 0$  such that

$$1 \leq \frac{\text{diam} T}{\rho_T} \leq C \quad \forall T \in \mathcal{T}_h, \forall h \in \mathcal{H}$$

$\rho_T$  is the diameter of the largest ball in  $T$ .

**Lemma 5.2.** Consider a shape regular conforming sequence of meshes  $\{\mathcal{T}_h\}_{h \in \mathcal{H}}$ . Then  $\exists$  a constant  $C_I > 0$  such that, for any  $u \in H^2(\Omega) \cap H_0^1(\Omega)$ , the interpolant  $I_h u \in V_h$  satisfies the bounds

$$\|u - I_h u\|_{H^1(\Omega)} \leq C_I h |u|_{H^2(\Omega)}$$

**Corollary 5.3.** If the equation has solution  $u \in H^2(\Omega) \cap H_0^1(\Omega)$ , then

$$\|u - u_h\|_{H^1} \leq C h |u|_{H^2(\Omega)}$$

which is similar to 1D linear rate of convergence for  $\|u - u_h\|_{H^1}$  with respect to mesh size.

**Theorem 5.4.** If  $\{\mathcal{T}_h\}_{h \in \mathcal{H}}$  is shape regular, then

$$\lim_{h \rightarrow 0} \|u - u_h\|_{H^1(\Omega)} \rightarrow 0$$

**Proof.** We aim to prove that  $\{V_h\}_{h \in \mathcal{H}}$  is dense in  $H_0^1(\Omega)$  in the limit, i.e.  $\lim_{h \rightarrow 0} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)} = 0$ ,  $\forall u \in H_0^1(\Omega)$ . Recall that  $H_0^1(\Omega)$  is the closure of  $C_0^\infty(\Omega)$  in  $H^1(\Omega)$  so  $\forall \varepsilon > 0$ ,  $\exists u_\varepsilon \in C_0^\infty(\Omega)$  such that  $\|u - u_\varepsilon\|_{H^1(\Omega)} < \varepsilon/2$ .

Since  $u_\varepsilon \in C_0^\infty(\Omega) \subset H^2(\Omega) \cap H_0^1(\Omega) \implies \|u_\varepsilon - I_h u_\varepsilon\| \leq C_I h |u_\varepsilon|_{H^2(\Omega)}$ . Then, for any  $h \leq \frac{\varepsilon}{2C_I |u_\varepsilon|_{H^2(\Omega)}} \implies \|u - I_h u_\varepsilon\|_{H^1(\Omega)} \leq \|u - u_\varepsilon\|_{H^1(\Omega)} + \|u_\varepsilon - I_h u_\varepsilon\|_{H^1(\Omega)} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2C_I |u_\varepsilon|_{H^2(\Omega)}} \leq \varepsilon$ . Therefore,  $\forall h > \frac{\varepsilon}{2C_I |u_\varepsilon|_{H^2(\Omega)}}$

$$\inf_{v_h \in V_h} \|u - v_h\|_{H^1} \leq \|u - I_h u_\varepsilon\|_{H^1} \leq \varepsilon$$

$$\implies \lim_{h \rightarrow 0} \inf \|u - v_h\|_{H^1} = 0 \implies \|u - u_h\|_{H^1} \rightarrow 0$$

as  $h \rightarrow 0$ . □

## 5.3 Other boundary conditions

**Example.** Let  $\Omega = (0, 1)^2$ , let  $\Gamma_N$  denote the right edge of  $\partial\Omega$  and

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \setminus \Gamma_N \\ \nabla u \cdot n &= 0 \text{ on } \Gamma_N \end{aligned}$$

Let  $H_{\Gamma_D}^1(\Omega) = \{v \in H^1(\Omega), v = 0 \text{ on } \Gamma_D = \partial\Omega \setminus \Gamma_N\}$ , let  $v \in H_{\Gamma_D}^1(\Omega)$  then

$$\int_{\Omega} (-\Delta u) v \, dx = \int_{\Omega} -\nabla \cdot (\nabla u) v \, dx$$

By divergence theorem,

$$\begin{aligned} \int_{\Omega} (-\Delta u) v \, dx &= \int_{\partial\Omega} -\nabla u \cdot n v \, dS + \int_{\Omega} \nabla u \cdot \nabla v \, dx \\ &= \int_{\Gamma_D} -\nabla u \cdot n v \, dS - \int_{\partial\Omega \setminus \Gamma_N} \nabla u \cdot n v \, dS + \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx \end{aligned}$$

The weak form is to find  $u \in H_{\Gamma_D}^1(\Omega)$  such that

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx = \ell(v) = \int_{\Omega} f v \, dx \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

**Exercise.** Show that we have Poincaré inequality for  $H_{\Gamma_D}^1(\Omega)$ :

$$\|v\|_{L^2(\Omega)} \leq C_p \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

then show that weak form has a unique solution in  $H_{\Gamma_D}^1(\Omega)$ .

Let  $x_{ij} = (ih_e, jh_e)$  where  $i, j = 0, \dots, k+1$ . Let  $\phi_{ij}$  be associated nodal hat functions for  $x_{ij}$ . Then  $\phi_{ij} \in H_{\Gamma_D}^1(\Omega)$  if and only if  $i \in \{1, \dots, k+1\}$  and  $j \in \{1, \dots, k\}$ . Therefore,  $V_h = \text{span}\{\phi_{ij}\}$ ,  $i \in \{1, \dots, k+1\}$  and  $j \in \{1, \dots, k\} \subset H_{\Gamma_D}^1(\Omega)$ . Error analysis then extend from Dirichlet case.

## 5.4 $L^2$ -norms of the error

**Definition 5.5.** The dual problem is

$$\begin{aligned} -\sum_{i,j=1}^2 \frac{\partial}{\partial x_i} (a_{ij} \frac{\partial z}{\partial x_j}) - \sum_{i=1}^2 \frac{\partial}{\partial x_i} (b_i z) + cz &= u - u_h \text{ in } \Omega \\ z &= 0 \text{ on } \partial\Omega \end{aligned}$$

We claim that the weak form of the equal above is to find  $z \in H_0^1(\Omega)$  such that

$$a(v, z) = \int_{\Omega} (u - u_h) v \, dx \quad \forall v \in H_0^1(\Omega)$$

Multiplying the equation by  $v \in H_0^1(\Omega)$  then

$$\begin{aligned} -\int_{\Omega} -\sum_{i,j=1}^2 \frac{\partial}{\partial x_i} (a_{ij} \frac{\partial z}{\partial x_j}) v \, dx &= \int_{\Omega} \sum_{i,j=1}^2 a_{ij} \frac{\partial z}{\partial x_j} \frac{\partial v}{\partial x_i} \, dx = \int_{\Omega} \sum_{i,j=1}^2 a_{ij} \frac{\partial z}{\partial x_i} \frac{\partial v}{\partial x_j} \, dx \\ \int_{\Omega} -\sum_{i,j=1}^2 \frac{\partial}{\partial x_i} (b_i z) v \, dx &= \int_{\Omega} \sum_{i,j=1}^2 b_i \frac{\partial v}{\partial x_i} = \int_{\Omega} \sum_{i=1}^2 b_i \frac{\partial v}{\partial x_i} z \, dx \end{aligned}$$

The bilinear form  $a(\cdot, \cdot)$  is coercive and bounded on  $H_0^1(\Omega)$ . Define  $\tilde{a}(w, v) = a(v, w)$ , then  $\tilde{\cdot}$  is also bounded and coercive on  $H_0^1(\Omega)$ , then  $\tilde{a}(z, v) = \int_{\Omega} (u - u_h) v \, dx$ . Apply LMT to deduce  $\exists! u \in H_0^1(\Omega)$  such that  $\tilde{a}(z, v) = \int_{\Omega} (u - u_h) v \, dx$ ,  $\forall v \in H_0^1(\Omega)$ .

We now assume that this problem has  $H^2$ -regularity which means for any  $u, u_h$  the solution  $z$  is in  $H^2(\Omega) \cap H_0^1(\Omega)$  and  $\exists C_R > 0$  such that  $|z|_{H^2(\Omega)} \leq C_R \|u - u_h\|_{L^2(\Omega)}$  where  $C_R$  independent of  $u, u_h$ .

**Example.** Consider

$$\begin{aligned} -u'' &= f \text{ in } (0, 1) \\ u(0) &= u(1) = 1 \end{aligned}$$

where the dual problem is

$$\begin{aligned} -z'' &= u - u_h \text{ in } (0, 1) \\ z(0) &= z(1) = 0 \end{aligned}$$

the dual problem is to find  $z \in H_0^1(\Omega)$  such that

$$\int_0^1 z' v' dx = \int_0^1 (u - u_h) v dx \quad \forall v \in H_0^1(\Omega)$$

and we have seen that  $z \in H^2(\Omega)$  then  $-z'' = -(u - u_h)$  then  $|z|_{H^1(0,1)} = \|z''\|_{L^2(0,1)} = \|u - u_h\|_{L^2(0,1)}$ .

**Example.** Consider  $\Omega \subset \mathbb{R}^2$  be open, bounded and convex then

$$\begin{aligned} \Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned}$$

Then the weak form is

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v dx = \int_{\Omega} f v dx \quad \forall v \in H_0^1(\Omega)$$

the dual problem is to find  $z \in H_0^1(\Omega)$  such that

$$\int_{\Omega} \nabla z \cdot \nabla v dx = \int_{\Omega} (u - u_h) v dx \quad \forall v \in H_0^1(\Omega)$$

since  $\Omega$  is convex so  $z \in H^2(\Omega)$  and  $|z|_{H^2(\Omega)} \leq C_{\Omega} \|u - u_h\|_{L^2(\Omega)}$  where  $C_{\Omega}$  depends on  $\Omega$  only.

**Theorem 5.6.** Suppose that the solution  $u$  of equation(2) is in  $H^2(\Omega) \cap H_0^1(\Omega)$ , also suppose the regularity on the dual problem. Suppose that  $\{\mathcal{T}_h\}_{h \in \mathcal{H}}$  is a shape regular sequence of meshes then  $\exists$  a constant  $C$  such that  $\|u - u_h\|_{L^2(0,1)} \leq Ch^2 |u|_{H^2(\Omega)}$ ,  $\forall h \in \mathcal{H}$ .

**Proof.** Since  $a(v, z) = \int_{\Omega} (u - u_h) v dx$ ,  $\forall v \in H_0^1(\Omega)$ . Pick  $v = u - u_h \in H_0^1(\Omega)$  then

$$\|u - u_h\|_{L^2(\Omega)}^2 = \int_{\Omega} |u - u_h|^2 dx = a(u - u_h, z)$$

Recall Galerkin's orthogonality  $a(u - u_h, v_h) = 0$ ,  $\forall v_h \in V_h$ . Pick  $v_h \in I_h z$  the interpolant of  $z$  into  $V_h$  then

$$\|u - u_h\|_{L^2(\Omega)}^2 = a(u - u_h, z - I_h z) \leq c_1 \|u - u_h\|_{H^1(\Omega)} \|z - I_h z\|_{H^1(\Omega)}$$

Since  $z \in H^2(\Omega)$ , we can use the bound for interpolant error  $\|z - I_h z\|_{H^1(\Omega)} \leq C_I h |z|_{H^2(\Omega)}$ . Therefore,

$$\begin{aligned} \|u - u_h\|_{L^2(\Omega)}^2 &\leq c_1 \|u - u_h\|_{H^1(\Omega)} C_I h |z|_{H^2(\Omega)} \\ \implies \|u - u_h\|_{L^2(\Omega)}^2 &\leq c_1 C_I C_R \|u - u_h\|_{L^2(\Omega)} \|u - u_h\|_{H^1(\Omega)} \\ \implies \|u - u_h\|_{L^2(\Omega)} &\leq c_1 C_I C_R h \|u - u_h\|_{H^1(\Omega)} \end{aligned}$$

Recall  $\|u - u_h\|_{H^1(\Omega)} \leq \frac{c_1}{c_0} C_I h |u|_{H^2(\Omega)}$  so

$$\|u - u_h\|_{L^2(\Omega)} \leq \frac{c_1^2 C_I^2 C_R}{c_0} h^2 |u|_{H^2(\Omega)}$$

□

**Example.** Consider

$$\begin{aligned} -u'' + u &= f \text{ in } (0, 1) \\ u(0) &= u(1) = 0 \end{aligned}$$

- Find WF, show existence and uniqueness of a solution.
- Show  $\|u - u_h\|_{H^1(\Omega)} \leq c_1 h \|f\|_{L^2(0,1)}$ ,  $\forall h$ .
- Show  $\|u - u_h\|_{L^2(0,1)} \leq C_2 h^2 \|f\|_{L^2(0,1)}$ .

(a) Multiply by  $v \in H_0^1(0, 1)$  and we need to find  $u \in H_0^1(\Omega)$  such that

$$a(u, v) = \int_0^1 u' v' + uv dx = \int_0^1 f v dx \quad \forall v \in H_0^1(0, 1)$$

To show the coercivity of the bilinear form

$$a(v, v) = \int_0^1 |v'|^2 + |v|^2 dx = \|v\|_{H^1(0,1)}^2$$

And for the boundedness,

$$|a(w, v)| \leq \|w\|_{H^1(0,1)} \|v\|_{H^1(0,1)}$$

so  $c_0 = c_1 = 1$ . And the boundedness of the functional is quite straightforward.

(b)  $a(u, v) = \ell(v)$ ,  $\forall v \in H_0^1(0, 1)$  then

$$\|u\|_{H^2(0,1)}^2 = a(u, u) = \ell(u) \leq \|f\|_{L^2(0,1)} \|u\|_{L^2(0,1)} \leq \|f\|_{L^2(0,1)} \|u\|_{H^1(0,1)}$$

the weak form is

$$\int_0^1 u' v' dx = \int_0^1 (f - u) v dx \quad \forall v \in H_0^1(0, 1)$$

Let  $\phi \in C_0^\infty(0, 1)$  then

$$\int_0^1 u \phi'' dx = - \int_0^1 u' \phi' dx = \int_\Omega (u - f) \phi dx \quad \forall \phi \in C_0^\infty(0, 1)$$

$\implies u - f$  satisfies the definition of second weak derivative of  $u$  so  $u \in H^2(0, 1)$  and  $\|u''\|_{L^2(0,1)} = \|u - f\|_{L^2(0,1)}$  then

$$\|u''\|_{L^2(0,1)} \leq \|u\|_{L^2(0,1)} + \|f\|_{L^2(0,1)} \leq 2\|f\|_{L^2(0,1)}$$

By Cea's Lemma

$$\begin{aligned} \|u - u_h\|_{H^1(0,1)} &\leq \frac{c_1}{c_0} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(0,1)} \leq \|u - I_h u\|_{H^1(0,1)} \leq \frac{h}{\pi} \left(1 + \frac{h^2}{\pi^2}\right)^{1/2} \|u''\|_{L^2(0,1)} \\ \implies \|u - u_h\|_{H^1(0,1)} &\leq \frac{2}{\pi} \left(1 + \frac{1}{\pi^2}\right)^{1/2} h \|f\|_{L^2(0,1)} \end{aligned}$$

(c) Define the dual problem

$$\begin{aligned} -z'' + z &= u - u_h \\ z(0) &= z(1) = 0 \end{aligned}$$

Find  $z \in H_0^1(0, 1)$ , such that

$$a(v, z) = \int_0^1 (u - u_h) z dx \quad \forall v \in H_0^1(0, 1)$$

Same analysis for  $u$  shows that  $z \in H^2(0, 1)$  and  $\|z''\|_{L^2(0,1)} \leq 2\|u - u_h\|_{L^2(0,1)}$ .

$$\begin{aligned} \implies \|u - u_h\|_{L^2(0,1)}^2 &= a(u - u_h, z) = a(u - u_h, z - I_h z) \\ a(u - u_h, z - I_h z) &\leq \|u - u_h\|_{H^1(0,1)} \|z - I_h z\|_{H^2(0,1)} \leq \|u - u_h\|_{H^1(0,1)} \frac{h}{\pi} \left(1 + \frac{1}{\pi^2}\right)^{1/2} \|z''\|_{L^2(0,1)} \\ &\leq \frac{2}{\pi} \left(1 + \frac{1}{\pi^2}\right)^{1/2} h \|u - u_h\|_{H^1(0,1)} \|u - u_h\|_{L^2(0,1)} \\ \implies \|u - u_h\|_{L^2(0,1)} &\leq \frac{2}{\pi} \left(1 + \frac{1}{\pi^2}\right)^{1/2} h \|u - u_h\|_{H^1(0,1)} \leq \left(\frac{2}{\pi} \left(1 + \frac{1}{\pi^2}\right)^{1/2}\right)^2 h^2 \|f\|_{L^2(0,1)} \end{aligned}$$

## 5.5 Higher-order approximation

So far we have treated finite element approximations using piecewise linear polynomials on a mesh. We can generate this to piecewise polynomial approximations of higher degrees.

**Example.** Consider

$$\begin{aligned} -u'' &= f \text{ in } (0, 1) \\ u(0) &= u(1) = 0 \end{aligned}$$

Let

$$\mathcal{T}_h = \{[x_{i-1}, x_i]\}_{i=1}^{N+1}$$

where  $0 = x_0 < x_1 < \dots < x_N < x_{N+1} = 1$ .  $h = x_i - x_{i-1} = \frac{1}{N+1}$ ,  $\forall i = 1, \dots, N+1$ . For each  $i = 1, \dots, N+1$ , let  $\psi_i(x)$  be defined by

$$\psi_i(x) = \begin{cases} \frac{1}{2}(x - x_{i-1})(x - x_i) & \forall x \in (x_{i-1}, x_i) \\ 0 & \text{otherwise} \end{cases}$$

Define  $V_{h,2} = \text{Span}\{\phi_i\}_{i=1}^N$  then  $\dim V_{h,2} = 2N + 1$ .

**Remark.**  $V_{h,2}$  is equivalently the space of all functions in  $H_0^1(0, 1)$  that are piecewise quadratic with respect to  $\mathcal{T}_h$ .

**Lemma 5.7.** There exists a linear map  $Q_h : H_0^1(0, 1) \cap H^2(0, 1) \rightarrow V_{h,2}$  that satisfies  $\|v - Q_h v\|_{L^2(x_{i-1}, x_i)} \leq \frac{h^3}{\pi^2} \|v'''\|_{L^2(x_{i-1}, x_i)}$  then

$$\|(v - Q_h v)'\|_{L^2(x_{i-1}, x_i)} \leq \frac{h^2}{\pi^2} \|v'''\|_{L^2(x_{i-1}, x_i)}$$

**Proof.** Let  $Q_h v(x) = \sum_{i=1}^N v(x_i) \phi_i(x) + \sum_{i=1}^{N+1} c_i \psi_i(x)$  where  $c_1 = \frac{1}{x_i - x_{i-1}} \int_{x_{i-1}}^{x_i} v'' ds$ . Define

$$\tilde{v} = v - \sum_{i=1}^{N+1} c_i \psi_i(x)$$

Recall the interpolant  $I_h : H_0^1(0, 1) \rightarrow V_h = \text{span}\{\phi_i\}_{i=1}^N$ .

$$v - Q_h v = v - \sum_{i=1}^N v(x_i) \phi_i(x) - \sum_{i=1}^{N+1} c_i \psi_i(x) = \tilde{v} - \sum_{i=1}^N v(x_i) \phi_i(x)$$

Since  $\psi_j(x) = 0, \forall j = 1, \dots, N+1 \implies \tilde{x}_i = v(x_i) = \sum_{j=1}^{N+1} c_j \psi_j(x_i) = v(x_j),$

$$\implies v - Q_h v = \tilde{v} - \sum_{i=1}^N \tilde{v} \phi_i(x) = \tilde{v} - I_h \tilde{v}$$

Bounds for  $I_h$  is

$$\|\tilde{v} - I_h \tilde{v}\|_{L^2(x_{i-1}, x_i)} \leq \frac{h^2}{\pi^2} \|\tilde{v}''\|_{L^2(x_{i-1}, x_i)}, \|\tilde{v} - I_h \tilde{v}\|'_{L^2(x_{i-1}, x_i)} \leq \frac{h}{\pi} \|\tilde{v}''\|_{L^2(x_{i-1}, x_i)}$$

Recall  $\tilde{v} = v - \sum_{i=1}^{N+1} c_i \psi_i \implies \tilde{v}|_{(x_{i-1}, x_i)} = v|_{(x_{i-1}, x_i)} - c_i \psi_i|_{(x_{i-1}, x_i)}$  then

$$\tilde{v}''|_{(x_{i-1}, x_i)} = v''|_{(x_{i-1}, x_i)} - c_i = v''|_{(x_{i-1}, x_i)} - \frac{1}{x_i - x_{i-1}} \int_{x_{i-1}}^{x_i} v'' dx$$

Poincaré inequality for functions with 0 mean value  $\|w\|_{L^2(a,b)} \leq (b-a)\|w'\|_{L^2(a,b)}$  if  $w \in H^1(a,b)$

$$\implies \|\tilde{v}''\|_{L^2(x_{i-1}, x_i)} \leq (x_i - x_{i-1}) \|v'''\|_{L^2(x_{i-1}, x_i)} = h \|v'''\|_{L^2(x_{i-1}, x_i)}$$

$$\implies \|v - Q_h v\|_{L^2(x_{i-1}, x_i)} \leq \frac{h^2}{\pi^2} \|\tilde{v}''\|_{L^2(x_{i-1}, x_i)} \leq \frac{h^3}{\pi^2} \|v'''\|_{L^2(x_{i-1}, x_i)}$$

□

Recall

$$\begin{aligned} -u'' &= f \text{ in } (0, 1) \\ u(0) &= u(1) = 0 \end{aligned}$$

Weak form is to find  $u \in H_0^1(0, 1)$  such that

$$\int_0^1 u' v' dx = \int_0^1 f v dx \quad \forall v \in H_0^1(0, 1)$$

FE is to find  $u_h \in V_{h,2}$  such that  $\int_0^1 u_h' v h' dx = \int_0^1 f v_h dx \quad \forall v_h \in V_{h,2}$ . By Céa's Lemma

$$\|u - u_h\|_{H^1(0,1)} \leq \frac{c_1}{c_0} \inf_{v_h \in V_{h,2}} \|u - v_h\|_{H^1(0,1)} \leq \frac{3}{2} \|u - Q_h u\|_{H^1(0,1)}$$

If  $u \in H^3(0, 1) \cap H_0^1(0, 1)$  then

$$\begin{aligned} \|u - Q_h u\|_{H^1(0,1)} &\leq \tilde{C} h^2 \|u'''\|_{L^2(0,1)} \\ \implies \|u - u_h\|_{H^1(0,1)} &\leq \frac{3}{2} \tilde{C} h^2 \|u'''\|_{L^2(0,1)} \end{aligned}$$